

# Supplementary Materials for “Distribution-Free Causal Inference for Semicontinuous Outcomes: A Two-Part Bayesian Causal Forest with Duan’s Smearing”

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**Abstract.** This supplementary document provides additional details, figures, and tables to support the main manuscript. Section 1 details the exact mathematical formulations of the data-generating processes (DGPs) under both link-linear and link-nonlinear specifications. Section 2 summarizes the MCMC algorithms, tree priors, hyperparameters, and diagnostic convergence checks. Section 3 presents the details of the computational complexity and timing analysis of the EDP+ZI benchmark model. Section 4 provides the complete set of simulation results, including the full 5-model sensitivity figures displaying the diverged Gamma benchmark lines. Section 5 is the self-contained oncology applied study, giving the outcome definitions, posterior analyses, model comparisons, literature positioning, and limitations for the SPECTRUM head-and-neck cancer trial that is summarized in the main manuscript. Section 6 is the self-contained health-economics applied study, in which the causal effect of dental-insurance coverage on annual dental expenditure is estimated from the 2023 Medical Expenditure Panel Survey and every estimate is reconciled against the applied dental health-economics literature.

## 1 Extended DGP Mathematical Specifications

In this section, the exact mathematical formulations of the three semicontinuous data-generating processes (DGP A, DGP B, and DGP C) evaluated in the simulation studies are presented, under both link-linear and link-nonlinear surfaces.

### 1.1 Covariates and Confounded Treatment Assignment

For all DGPs, a baseline covariate matrix  $X_i \in \mathbb{R}^P$  with  $P = 5$  dimensions is generated, where each covariate is drawn independently from a standard normal distribution:

$$X_{ip} \sim \mathcal{N}(0, 1), \quad \text{for } p = 1, \dots, 5 \quad (1.1)$$

To simulate observational study conditions with strong selection and confounding, the treatment propensity score  $\pi(X_i) = P(Z_i = 1 \mid X_i)$  is designed to be nonlinear in  $X_{i2}$ :

$$\pi(X_i) = \Phi(-0.5 + 0.4X_{i1} + 0.3X_{i2}^2) \quad (1.2)$$

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where  $\Phi(\cdot)$  is the standard normal cumulative distribution function (CDF). Binary treatment assignments are generated via:

$$Z_i \sim \text{Bernoulli}(\pi(X_i)) \quad (1.3)$$

### 1.2 Link-Linear Data-Generating Processes

The link-linear DGPs represent the classical benchmark design where conditional means are linear functions of the covariates on their respective link scales. Let  $c_i$  be the intercept parameter used for zero-inflation calibration.

#### DGP A: Log-Normal Hurdle DGP

The binary hurdle indicator  $I_i = \mathbb{I}(Y_i > 0)$  is generated via:

$$I_i \sim \text{Bernoulli}(p_i) \quad (1.4)$$

$$p_i = \Phi(c_i + 0.5X_{i1} - 0.3X_{i3} + Z_i(0.2 + 0.1X_{i1})) \quad (1.5)$$

The continuous positive intensity  $Y_i^+ = (Y_i \mid I_i = 1)$  follows a log-normal distribution:

$$\log(Y_i^+) \sim \mathcal{N}(1.5 + 0.8X_{i2} + 0.4X_{i4} + Z_i(0.25 - 0.15X_{i2}), \sigma^2) \quad (1.6)$$

where  $\sigma = 0.5$  (so the residual log-scale variance is  $\sigma^2 = 0.25$ ). The observed outcome is  $Y_i = I_i \cdot Y_i^+$ . The true response-scale potential outcomes are:

$$\mu_0(X_i) = \Phi(c_i + 0.5X_{i1} - 0.3X_{i3}) \cdot \exp(1.625 + 0.8X_{i2} + 0.4X_{i4}) \quad (1.7)$$

$$\mu_1(X_i) = \Phi(c_i + 0.5X_{i1} - 0.3X_{i3} + 0.2 + 0.1X_{i1}) \cdot \exp(1.875 + 0.65X_{i2} + 0.4X_{i4}) \quad (1.8)$$

where the addition of  $0.5\sigma^2 = 0.125$  inside the exponent reflects the log-normal expectation correction. The hurdle intercept is set to  $c_i = 0.2$ , calibrating the marginal zero proportion to  $\approx 37\%$  zeros.

#### DGP B: Gamma Hurdle DGP

The binary hurdle indicator is identical in structure to DGP A. The continuous positive outcome intensity  $Y_i^+ = (Y_i \mid I_i = 1)$  is generated from a Gamma distribution:

$$Y_i^+ \sim \text{Gamma}(\text{shape} = \alpha, \text{scale} = \mu_{c,i}/\alpha) \quad \text{with } \alpha = 2.0 \quad (1.9)$$

where the conditional log-mean is given by:

$$\log(\mu_{c,i}) = 1.5 + 0.8X_{i2} + 0.4X_{i4} + Z_i(0.25 - 0.15X_{i2}) \quad (1.10)$$

The true response-scale potential outcomes are:

$$\mu_0(X_i) = \Phi(c_i + 0.5X_{i1} - 0.3X_{i3}) \cdot \exp(1.5 + 0.8X_{i2} + 0.4X_{i4}) \quad (1.11)$$

$$\mu_1(X_i) = \Phi(c_i + 0.5X_{i1} - 0.3X_{i3} + 0.2 + 0.1X_{i1}) \cdot \exp(1.75 + 0.65X_{i2} + 0.4X_{i4}) \quad (1.12)$$

The hurdle intercept is set to  $c_i = 0.2$ , calibrating standard zero-inflation to  $\approx 39.5\%$  zeros.

**DGP C: Tweedie Semicontinuous DGP**

The outcome  $Y_i$  is generated directly from an intrinsic compound Poisson-Gamma Tweedie dispersion model, which generates zeros and positive values jointly:

$$Y_i \sim \text{Tweedie}(\mu_i, \phi = 1.5, p = 1.5) \quad (1.13)$$

where the mean parameter  $\mu_i$  is governed by the log link:

$$\log \mu_i = 1.2 + c_i + 0.8X_{i1} - 0.4X_{i3} + Z_i(0.3 + 0.15X_{i1}) \quad (1.14)$$

The true potential outcomes and CATE on the raw response scale are:

$$\mu_0(X_i) = \exp(1.2 + c_i + 0.8X_{i1} - 0.4X_{i3}) \quad (1.15)$$

$$\mu_1(X_i) = \exp(1.5 + c_i + 0.95X_{i1} - 0.4X_{i3}) \quad (1.16)$$

Setting  $c_i = 0.0$  calibrates nominal zero proportion to  $\approx 50\%$  zeros.

**1.3 Link-Nonlinear Data-Generating Processes**

To stress-test the estimators under functional misspecification, the affine predictors are replaced with mean-centered nonlinear functions, covariate interactions, and bounded treatment effects. Four shared nonlinear building blocks are defined:

$$h(X_i) = 0.50 \sin(1.3X_{i1}) - 0.30X_{i3} + 0.35X_{i1}X_{i2} - 0.20(X_{i3}^2 - 1) \quad (1.17)$$

$$\tau_h(X_i) = 0.20 + 0.20X_{i1}X_{i2} \quad (1.18)$$

$$m(X_i) = 1.50 + 0.80 \sin(1.2X_{i2}) + 0.40X_{i4} + 0.35X_{i2}X_{i4} - 0.25(X_{i1}^2 - 1) \quad (1.19)$$

$$\tau_c(X_i) = 0.25 - 0.15 \sin(1.3X_{i2}) + 0.12 \tanh(X_{i1}X_{i2}) \quad (1.20)$$

Here, the terms  $X_{ij}^2 - 1$  and the sine/interaction functions are designed to have an expectation of approximately 0 under  $X \sim \mathcal{N}(0, I_5)$ , keeping the outcome scale and zero proportion comparable to the linear settings.

**DGP A: Nonlinear Log-Normal Hurdle DGP**

The hurdle potential outcome probabilities are defined as:

$$p_0(X_i) = \Phi(c_i + h(X_i)) \quad (1.21)$$

$$p_1(X_i) = \Phi(c_i + h(X_i) + \tau_h(X_i)) \quad (1.22)$$

The continuous positive intensity  $Y_i^+ = (Y_i \mid I_i = 1)$  is generated as:

$$\log(Y_i^+) \sim \mathcal{N}(m(X_i) + Z_i\tau_c(X_i), \sigma^2) \quad (1.23)$$

with  $\sigma = 0.5$  ( $\sigma^2 = 0.25$ ). The true potential outcomes are:

$$\mu_0(X_i) = p_0(X_i) \cdot \exp(m(X_i) + 0.125) \quad (1.24)$$

$$\mu_1(X_i) = p_1(X_i) \cdot \exp(m(X_i) + \tau_c(X_i) + 0.125) \quad (1.25)$$

The intercept is set to  $c_i = 0.2$ , yielding a standard zero proportion of  $\approx 40\%$ .

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### DGP B: Nonlinear Gamma Hurdle DGP

The hurdle probabilities are identical to nonlinear DGP A. The continuous positive intensity  $Y_i^+ = (Y_i \mid I_i = 1)$  is generated via:

$$Y_i^+ \sim \text{Gamma}(\text{shape} = \alpha, \text{scale} = \mu_{c,i}(Z_i)/\alpha) \quad \text{with } \alpha = 2.0 \quad (1.26)$$

where the conditional means are:

$$\mu_{c,i}(0) = \exp(m(X_i)) \quad (1.27)$$

$$\mu_{c,i}(1) = \exp(m(X_i) + \tau_c(X_i)) \quad (1.28)$$

The true potential outcomes are:

$$\mu_0(X_i) = p_0(X_i) \cdot \exp(m(X_i)) \quad (1.29)$$

$$\mu_1(X_i) = p_1(X_i) \cdot \exp(m(X_i) + \tau_c(X_i)) \quad (1.30)$$

Setting  $c_i = 0.2$  yields a standard zero proportion of  $\approx 40\%$ .

### DGP C: Nonlinear Tweedie DGP

The Tweedie parameters are  $\phi = 1.5, p = 1.5$ . The nonlinear log-mean function and its treatment effect component are defined as:

$$g_c(X_i) = 1.2 + 0.80 \sin(1.2X_{i1}) - 0.40X_{i3} + 0.35X_{i1}X_{i3} - 0.25(X_{i2}^2 - 1) \quad (1.31)$$

$$\tau_{\text{Tweedie}}(X_i) = 0.30 - 0.15 \sin(1.3X_{i1}) + 0.12 \tanh(X_{i1}X_{i3}) \quad (1.32)$$

The conditional Tweedie mean is  $\log \mu_i = c_i + g_c(X_i) + Z_i \tau_{\text{Tweedie}}(X_i)$ . The true potential outcomes are:

$$\mu_0(X_i) = \exp(c_i + g_c(X_i)) \quad (1.33)$$

$$\mu_1(X_i) = \exp(c_i + g_c(X_i) + \tau_{\text{Tweedie}}(X_i)) \quad (1.34)$$

Setting the intercept shift to  $c_i = -2.072$  calibrates the standard zero proportion to  $\approx 40\%$  (harmonizing it with DGP A and B).

## 2 MCMC Details & Diagnostics

### 2.1 Proof of Lemma 3.3 (Identification on the Active Subpopulation)

The main manuscript states Lemma 3.3 in compressed form. The full statement and its proof are given here.

**Lemma 3.3** (Identification on the active subpopulation). Let  $L_i(z) = \log Y_i^+(z)$  for  $z \in \{0, 1\}$  and define the active-subpopulation log-scale contrast

$$\tau_c^I(x) = E[L_i(1) - L_i(0) \mid X_i = x, I_i = 1]. \quad (2.1)$$

Suppose that (i) Assumption 3.1 of the main manuscript holds, (ii) consistency holds on the active subpopulation, that is,  $\log Y_i^+ = L_i(Z_i)$  whenever  $I_i = 1$ , and (iii) there exists a constant  $\delta_+ > 0$  such that

$$\delta_+ < \pi^+(x) = P(Z_i = 1 \mid X_i = x, I_i = 1) < 1 - \delta_+ \quad (2.2)$$

for every  $x$  in the support of  $X_i \mid I_i = 1$ . Then

$$\tau_c^I(x) = E[\log Y_i^+ \mid X_i = x, Z_i = 1, I_i = 1] - E[\log Y_i^+ \mid X_i = x, Z_i = 0, I_i = 1], \quad (2.3)$$

so  $\tau_c^I(x)$  is identified from the observed active-subpopulation distribution, and under the continuous-stage conditional mean model the treatment function  $\tau_c(x)$  represents this identified contrast.

*Proof.* Fix  $z \in \{0, 1\}$  and  $x$  in the support of  $X_i \mid I_i = 1$ . Consistency on the active subpopulation replaces the observed log outcome by the corresponding potential outcome, and Assumption 3.1 then removes the conditioning on  $Z_i$ :

$$E[\log Y_i^+ \mid X_i = x, Z_i = z, I_i = 1] = E[L_i(z) \mid X_i = x, Z_i = z, I_i = 1] = E[L_i(z) \mid X_i = x, I_i = 1]. \quad (2.4)$$

Condition (iii) guarantees that both arms retain positive conditional probability at every such  $x$ , so both observed conditional means on the left-hand side are well defined. Subtracting the display for  $z = 0$  from the display for  $z = 1$  gives (2.3).  $\square$

**Remark.** Applying the propensity-score balancing argument of [Rosenbaum and Rubin \(1983\)](#) to the conditional distribution given  $I_i = 1$  yields  $(Y_i^+(0), Y_i^+(1)) \perp\!\!\!\perp Z_i \mid \pi^+(X_i), I_i = 1$ , so  $\pi^+(X_i)$  is a balancing score within the active subpopulation. Because  $\pi^+(X_i)$  is a function of  $X_i$ , however, supplying it to the prognostic forest  $\mu_c(\cdot)$  changes the model's regularization without adding any identifying condition beyond (i)–(iii). In particular it does not substitute for Assumption 3.1, for active-subpopulation overlap, or for adjustment on  $X_i$  itself when the target contrast is heterogeneous in  $X_i$ .

## 2.2 Posterior Sampling Details

The posterior distribution of the ZIC-BCF-Smear model is obtained from a single Gibbs sampler in which the hurdle block and the intensity block are updated in turn within each sweep, and are written to a common draw index after burn-in. The two blocks are as follows.

1. **Hurdle Stage (Probit BCF):** Let  $I_i = \mathbb{I}(Y_i > 0)$  be the binary response indicator. Latent Gaussian utility variables  $I_i^* \sim \mathcal{N}(\mu_b(X_i, \hat{\pi}_i) + Z_i \tau_b(X_i), 1)$  are introduced such that  $I_i = \mathbb{I}(I_i^* > 0)$ . Using the data augmentation algorithm of [Albert and Chib \(1993\)](#), the latent variables  $I_i^*$  are sampled from truncated normal distributions:

$$(I_i^* \mid I_i = 1) \sim \mathcal{N}(\mu_b(X_i, \hat{\pi}_i) + Z_i \tau_b(X_i), 1) \text{ truncated to } (0, \infty) \quad (2.5)$$

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$$(I_i^* \mid I_i = 0) \sim \mathcal{N}(\mu_b(X_i, \hat{\pi}_i) + Z_i \tau_b(X_i), 1) \text{ truncated to } (-\infty, 0] \quad (2.6)$$

Given the latent outcomes  $I_i^*$ , the regression forests  $\mu_b(\cdot)$  and  $\tau_b(\cdot)$  are updated using standard Bayesian backfitting updates for BART (Chipman et al., 2010).

2. **Continuous Stage (Gaussian BCF with SPA):** On the active subpopulation  $\{i \mid Y_i > 0\}$ , the log outcomes  $y_{i,\text{active}} = \log(Y_i^+)$  are computed and standardized to  $y_{i,\text{active}}^*$ . The continuous BCF model is fit on these standardized outcomes, conditioning on the active-set propensity score  $\hat{\pi}_i^+$ . Updates for the forests  $\mu_c(\cdot)$ ,  $\tau_c(\cdot)$ , and the residual variance  $\sigma_c^2$  are executed using Gaussian backfitting steps.

### Exactness of the Block-Wise Sweep

Updating the two blocks one after the other within a sweep, rather than through a joint proposal, is not an approximation. The two-part likelihood factorizes exactly,

$$p(\{Y_i\} \mid \theta_b, \theta_c) = \underbrace{\prod_{i=1}^N \Phi(\mu_{bi} + Z_i \tau_{bi})^{I_i} [1 - \Phi(\mu_{bi} + Z_i \tau_{bi})]^{1-I_i}}_{\text{participation, all } N \text{ units}} \times \underbrace{\prod_{i:I_i=1} p(Y_i^+ \mid \theta_c)}_{\text{intensity, active units only}} \quad (2.7)$$

because the participation indicators  $\{I_i\}$  are a deterministic function of the observed data and the two blocks share no parameter. Combined with independent BART priors on  $\theta_b = (\mu_b, \tau_b)$  and  $\theta_c = (\mu_c, \tau_c, \sigma_c^2)$ , (2.7) gives  $p(\theta_b, \theta_c \mid \text{data}) = p(\theta_b \mid \text{data}) p(\theta_c \mid \text{data})$ . The blocks are therefore *a posteriori* independent, the block-wise sweep is an exact Gibbs sampler for the joint posterior, and nothing is lost relative to a hypothetical joint update.

Two practical consequences follow. First, there is nothing to gain from coupling the blocks more tightly than the sampler already does. Second, there is something to lose from decoupling them further: running two separate chains and pairing their draws arbitrarily would be valid only because of the same factorization, while forfeiting the shared random state and the shared convergence diagnostics that make the composite functional  $\tau_{\text{resp}}(X_i)$ , a product of a participation probability and an intensity, straightforward to summarize.

The one place where an approximation genuinely enters is the propensity scores. The full-sample  $\hat{\pi}_i$  and the subpopulation  $\hat{\pi}_i^+$  are estimated once by logistic regression before the sampler starts and are then held fixed as covariates, so their estimation uncertainty is not propagated into the outcome posterior. This is a deliberate cut, and it is the same cut made by BCF itself (Hahn et al., 2020): feeding the propensity score back into a joint posterior would let the flexible outcome model distort the treatment model, and the plug-in estimate is what makes  $\hat{\pi}_i$  a stable basis function for the prognostic ensemble rather than a parameter competing with it. The cost is a mild understatement of posterior uncertainty in settings with weak overlap, where the propensity model is itself poorly determined.

## 2.3 Complete Model Fitting Algorithm

For reference, the full ZIC-BCF-Smear fitting and re-transformation procedure summarized in the main manuscript is stated below as pseudocode.

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### Algorithm 1: ZIC-BCF-Smear MCMC and Smearing Retransformation

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**Input:** Semicontinuous outcomes  $Y_i \geq 0$ , binary treatments  $Z_i \in \{0, 1\}$ , design matrices  $X_i$ .

**Output:** Posterior draws of response-scale potential outcomes  $\mu_0^{(s)}(X_i)$ ,  $\mu_1^{(s)}(X_i)$ , and CATE  $\tau_{\text{resp}}^{(s)}(X_i)$ .

1. Fit the full-sample propensity score  $\hat{\pi}_i = P(Z_i = 1 \mid X_i)$ .
2. Identify the active subpopulation  $\{i : Y_i > 0\}$ . Fit the subpopulation propensity score  $\hat{\pi}_i^+ = P(Z_i = 1 \mid X_i, Y_i > 0)$  on this subset.
3. Run the Probit BCF model on the full sample to obtain  $S$  posterior MCMC draws of the latent hurdle functions  $\mu_{bi}^{(s)}$  and  $\tau_{bi}^{(s)}$ .
4. Run the Gaussian BCF model on the active subset for the log-outcomes  $\log(Y_i^+)$  to obtain  $S$  posterior MCMC draws of the continuous intensity functions  $\mu_{ci}^{(s)}$ ,  $\tau_{ci}^{(s)}$ , and log-residual variance  $\sigma_c^{(s)}$ .
5. For each saved posterior MCMC draw  $s \in \{1, \dots, S\}$ :
  - (a) Compute active log-scale residuals:  $e_j^{(s)} = \log(Y_j^+) - [\mu_{cj}^{(s)} + Z_j \tau_{cj}^{(s)}]$  for all  $j \in \text{active}$ .
  - (b) Compute the smearing factor:  $\phi_{\text{smear}}^{(s)} = \frac{1}{N_{\text{active}}} \sum_{j \in \text{active}} \exp(e_j^{(s)})$ .
  - (c) Re-transform predictions for all  $N$  units to the response scale:

$$\begin{aligned}\mu_0^{(s)}(X_i) &= \Phi(\mu_{bi}^{(s)}) \cdot \exp(\mu_{ci}^{(s)}) \cdot \phi_{\text{smear}}^{(s)} \\ \mu_1^{(s)}(X_i) &= \Phi(\mu_{bi}^{(s)} + \tau_{bi}^{(s)}) \cdot \exp(\mu_{ci}^{(s)} + \tau_{ci}^{(s)}) \cdot \phi_{\text{smear}}^{(s)}\end{aligned}$$

- (d) Compute CATE draws:  $\tau_{\text{resp}}^{(s)}(X_i) = \mu_1^{(s)}(X_i) - \mu_0^{(s)}(X_i)$ .
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## 2.4 Tree Prior Hyperparameter Choices

Hyperparameters are set to standard default values in the literature ([Hahn et al., 2020](#)).

- **Forest Sizes:** The prognostic forests  $(\mu_b, \mu_c)$  are allocated  $T_{\text{control}} = 200$  trees. The treatment-moderating forests  $(\tau_b, \tau_c)$  are allocated  $T_{\text{moderate}} = 50$  trees, regularizing treatment effects to prevent overfitting.
- **Tree Structure Prior:** For any node at depth  $d$ , the probability of splitting is given by  $\alpha(1 + d)^{-\beta}$ . The settings are:
  - Prognostic forests:  $\alpha_c = 0.95, \beta_c = 2.0$ .

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- Moderating forests:  $\alpha_m = 0.25, \beta_m = 3.0$ .

The much smaller  $\alpha_m$  and larger  $\beta_m$  heavily penalize deep trees for the treatment effect, reflecting the prior that treatment effect heterogeneity is smooth and low-dimensional.

- **Leaf Parameters Scale Prior:** The leaf parameters are assigned normal priors  $\mathcal{N}(0, \sigma_{\text{leaf}}^2)$ .
  - Hurdle stage: the prior medians of the cumulative forest scales are set to  $\sigma_{\mu_b} = 2.0$  (latent scale) and  $\sigma_{\tau_b} = 1.0$ .
  - Continuous stage: prior medians are scaled data-adaptively:  $\sigma_{\mu_c} = 2.0 \cdot \text{sd}(\log(Y_{\text{active}}))$  and  $\sigma_{\tau_c} = 0.25 \cdot \text{sd}(\log(Y_{\text{active}})) / \text{sd}(Z_{\text{active}})$ .
- **Variance Prior:** The continuous error variance is assigned a conjugate scaled inverse chi-squared prior  $\sigma_c^2 \sim \text{Inv-}\chi^2(\nu, \lambda)$  with degrees of freedom  $\nu = 3$ . The scale parameter  $\lambda$  is calibrated such that the prior 90th percentile of  $\sigma_c$  matches the residual standard deviation from a linear model fit on the active subset.

## 2.5 MCMC Convergence Diagnostics

Convergence is assessed rather than assumed, and the assessment is part of the software rather than a separate manual exercise. The `zicbcf` package exports `zicbcf_convergence()`, which takes one or more fitted chains and returns, for every monitored functional, the Gelman–Rubin potential scale reduction factor  $\hat{R}$  (Gelman and Rubin, 1992), the effective sample size, the Monte Carlo standard error, and the Geweke spectral-density statistic comparing the first tenth of each chain with its last half (Geweke, 1992). All three are obtained from `coda` (Plummer et al., 2006). When a single chain is supplied,  $\hat{R}$  is the split-chain variant, so within-chain drift is still detectable; when several chains are supplied it is the split version of the across-chain statistic. A companion function, `zicbcf_convergence_table()`, applies the same three diagnostics to any set of posterior draws, and is used for the Gamma-hurdle benchmark, whose fits are not `zicbcf` objects.

The monitored set is deliberately not restricted to the average treatment effect, because a two-part model can mix well on an aggregate while mixing poorly on a component. It comprises 1. the response-scale ATE, 2. the participation-margin ATE, 3. Duan’s smearing factor, 4. the log-scale residual standard deviation  $\sigma_c$ , 5. the cross-unit means of the four forest surfaces  $\mu_b$ ,  $\tau_b$ ,  $\mu_c$ , and  $\tau_c$ , and 6. ten unit-level CATEs chosen at evenly spaced quantiles of the posterior-mean effect distribution, so that the monitored units span the estimated effect surface rather than an arbitrary corner of it.

### Simulation Grid

Diagnostics for the simulation study are produced by `simulation_studies/run_convergence_diagnostics.R`, which fits four independent chains of 1,000 retained draws after 1,000 burn-in iterations to five replicates of each of the six data-generating processes at

$N = 500$ , and writes the per-quantity table together with a per-DGP summary. Table 1 reports the summary.

Table 1: Convergence diagnostics for the simulation grid: four chains of 1,000 retained draws on each of five replicates per data-generating process at  $N = 500$ . Global summaries are the eight scalar functionals listed above; unit-level CATEs are ten units spanning the effect surface. The Geweke column is the largest absolute value over all chains and quantities in the cell.

| Data-generating process             | Monitored set    | Median $\hat{R}$ | 95th pct. $\hat{R}$ | Max $\hat{R}$ | Median ESS | Min ESS | Max $ z_{\text{Geweke}} $ |
|-------------------------------------|------------------|------------------|---------------------|---------------|------------|---------|---------------------------|
| Linear DGP A (log-normal hurdle)    | Global summaries | 1.004            | 1.014               | 1.017         | 928        | 449     | 4.05                      |
|                                     | Unit-level CATEs | 1.005            | 1.012               | 1.048         | 1,125      | 508     | 3.68                      |
| Linear DGP B (Gamma hurdle)         | Global summaries | 1.002            | 1.007               | 1.010         | 1,124      | 444     | 2.57                      |
|                                     | Unit-level CATEs | 1.003            | 1.010               | 1.022         | 1,563      | 693     | 3.53                      |
| Linear DGP C (Tweedie)              | Global summaries | 1.004            | 1.038               | 1.118         | 1,113      | 203     | 4.08                      |
|                                     | Unit-level CATEs | 1.004            | 1.011               | 1.030         | 1,489      | 869     | 4.91                      |
| Nonlinear DGP A (log-normal hurdle) | Global summaries | 1.008            | 1.050               | 1.131         | 668        | 159     | 7.06                      |
|                                     | Unit-level CATEs | 1.009            | 1.027               | 1.074         | 890        | 396     | 4.99                      |
| Nonlinear DGP B (Gamma hurdle)      | Global summaries | 1.003            | 1.012               | 1.015         | 888        | 451     | 4.20                      |
|                                     | Unit-level CATEs | 1.006            | 1.015               | 1.026         | 1,398      | 532     | 4.13                      |
| Nonlinear DGP C (Tweedie)           | Global summaries | 1.005            | 1.074               | 1.162         | 1,086      | 161     | 4.46                      |
|                                     | Unit-level CATEs | 1.004            | 1.009               | 1.016         | 1,646      | 966     | 3.43                      |

The picture Table 1 gives is not uniformly clean, and it should not be reported as though it were. Median  $\hat{R}$  never exceeds 1.009 in any cell, and in the three link-linear cells and the nonlinear Gamma cell the 95th percentile stays below 1.015. The two hardest cells, nonlinear DGP A and nonlinear DGP C, carry a small tail of poorly mixed quantities: the 95th percentile of  $\hat{R}$  reaches 1.050 and 1.074 and the maximum reaches 1.131 and 1.162, with effective sample sizes as low as 159 out of 4,000 pooled draws. Those values attach to global forest-scale summaries rather than to unit-level effects, and they occur in exactly the link-nonlinear configurations that the main manuscript identifies as most demanding. The practical reading is that a budget of 1,000 retained draws is adequate for the reported Monte Carlo summaries, whose accuracy depends on averaging over 100 replicates rather than on any single chain, but is not generous for a single applied fit. That is why both applied studies use four chains and a larger per-chain budget, and report  $\hat{R}$  below 1.01 throughout.

### Applied Studies

For the final SPECTRUM fits, the summed-duration endpoint has maximum  $\hat{R} = 1.001$ , minimum effective sample size 2,670, and maximum absolute Geweke statistic 2.03 for ZIC-BCF-Smear, while the Gamma hurdle has corresponding values 1.000, 16,000, and 1.42. On the distinct-calendar-day endpoint, the corresponding values are 1.002, 2,777, and 2.21 for ZIC-BCF-Smear and 1.000, 16,000, and 1.42 for the Gamma hurdle. All of these sit well inside the calibrated null ranges at the multiplicity that the monitored set implies, and the final fits use the 5,000-iteration burn-in motivated in Section 5.4 (the originally reported 2,000-iteration configuration produced largest statistics of 3.67 on the summed-duration endpoint and 4.90 on the distinct-calendar-day endpoint, both on  $\sigma_c$ ). The MEPS fits use the same 5,000-iteration burn-in, and their diagnostics are reported below.

## 2.6 Homoskedasticity of the Log-Scale Errors

Duan’s re-transformation replaces the parametric multiplier  $\exp(\sigma_c^2/2)$  with the non-parametric factor

$$\hat{\phi}^{(s)} = \frac{1}{n_+} \sum_{i \in \mathcal{A}} \exp\left(\hat{\varepsilon}_i^{(s)}\right), \quad \hat{\varepsilon}_i^{(s)} = \log Y_i - \mu_c^{(s)}(X_i) - Z_i \tau_c^{(s)}(X_i), \quad (2.8)$$

one scalar per retained draw, applied identically to both counterfactual arms. Equation (2.8) is consistent for the response-scale mean when the log-scale errors are independent of the covariates. Under heteroskedasticity the correct multiplier varies with  $x$ , and because both potential-outcome means are multiplied by the same scalar, every response-scale estimate scales linearly in it. The assumption is therefore consequential and is tested.

The package exports `zicbcf_smearing_diagnostics()`, which returns three things. First, a studentized Breusch–Pagan test (Breusch and Pagan, 1979; Koenker, 1981) regressing the squared log-scale residuals on the fitted log-scale mean and on the covariates; the studentized form is used because the raw Breusch–Pagan statistic assumes normal errors, which is precisely what a smearing re-transformation exists to avoid assuming. Second, the same test recomputed inside each of a thinned set of retained draws, so the evidence is summarized by a posterior distribution of  $p$ -values rather than by a single plug-in number. Third, and most useful in practice, a locally smeared sensitivity analysis that replaces the single scalar with stratum-specific factors over quantiles of the fitted log-scale mean and recomputes the average treatment effect on that basis. If the locally smeared estimate is close to the scalar-smeared one, the assumption is not materially driving the reported effect, whatever the test says about it.

Table 2 applies this to the simulation grid, where the truth is known. DGP A generates log-normal errors with a constant log-scale variance and therefore acts as a negative control; the Gamma and Tweedie generators do not, and act as positive ones.

Table 2: Homoskedasticity of the log-scale errors across the simulation grid, five replicates per data-generating process at  $N = 500$ . The residual-scale ratio is the largest divided by the smallest residual standard deviation across five strata of the fitted log-scale mean. The relative change is the locally smeared average treatment effect divided by the scalar-smeared one, minus one.

| Data-generating process             | Median BP $p$        | Rejected at 5% | Residual-scale ratio | Median relative change | Max  relative change |
|-------------------------------------|----------------------|----------------|----------------------|------------------------|----------------------|
| Linear DGP A (log-normal hurdle)    | 0.194                | 1 of 5         | 1.23                 | +0.9%                  | 3.1%                 |
| Linear DGP B (Gamma hurdle)         | 0.026                | 3 of 5         | 1.21                 | −4.4%                  | 8.5%                 |
| Linear DGP C (Tweedie)              | $2.9 \times 10^{-7}$ | 5 of 5         | 2.12                 | −17.7%                 | 19.5%                |
| Nonlinear DGP A (log-normal hurdle) | 0.313                | 0 of 5         | 1.21                 | +1.3%                  | 2.6%                 |
| Nonlinear DGP B (Gamma hurdle)      | 0.242                | 1 of 5         | 1.27                 | +2.5%                  | 7.7%                 |
| Nonlinear DGP C (Tweedie)           | $7.0 \times 10^{-4}$ | 5 of 5         | 1.54                 | −25.1%                 | 32.5%                |

The pattern is the one the diagnostic is meant to produce. Where homoskedasticity holds by construction the test does not reject and the locally smeared estimate is within a few percent of the scalar-smeared one. Where it fails, the test rejects decisively and the locally smeared estimate moves materially, which is the quantity a reader actually needs in order to judge how much the assumption is doing. The applied results are reported with their studies, in Sections 5 and 6.

### 3 EDP+ZI Timing Analysis

The Enriched Dirichlet Process Zero-Inflated (EDP+ZI) model (Kim et al., 2024) is a flexible Bayesian nonparametric benchmark that models semicontinuous data by placing a Dirichlet process mixture prior over the joint distribution of hurdle and continuous intensity coefficients. However, its high computational complexity limits its practical utility for large datasets or simulation studies.

#### 3.1 Computational Bottleneck of Dirichlet Process Mixtures

In EDP+ZI, observations are dynamically grouped into latent clusters. At each Gibbs sweep, the cluster assignments and the cluster-specific regression parameters are updated. For a dataset of size  $N$ , let  $N_k$  be the size of cluster  $k$ . Updating the multivariate regression parameters for cluster  $k$  requires inverting the covariance matrix of the design matrix, which has dimension  $P \times P$  (where  $P$  is the number of covariates). While this step is relatively fast, updating cluster assignments and evaluating marginal likelihoods for each observation involves computing multivariate normal densities that scale with the cluster sample size. In more general formulations (or when handling latent variables for zero-inflated outcomes), Gibbs updates require matrix operations that scale cubically with the sample size,  $O(N^3)$ , due to joint multivariate Normal updates or Gaussian Process approximations.

In contrast, BCF models represent nonparametric functions using sum-of-trees. The computational cost of fitting a single regression tree of depth  $d$  on a dataset of size  $N$  with  $P$  covariates scales as  $O(dPN \log N)$ , which is dominated by sorting the covariates at each split. For a forest of  $T$  trees, the computational complexity per MCMC sweep is:

$$\mathcal{O}(T \cdot P \cdot N \log N) \quad (3.1)$$

This linear-logarithmic scaling allows BCF models to scale efficiently to very large datasets, whereas the cubic cost of EDP+ZI becomes prohibitive.

#### 3.2 Empirical Timing Comparisons

The EDP+ZI model was implemented and verified using the authors' reference code. The empirical runtimes on standard CPU threads are summarized below:

- **ZIC-BCF-Smear** ( $N = 500$ ): A single simulation run (1,000 burn-in + 1,000 sampling iterations) takes approximately **18 minutes** ( $\approx 0.3$  hours). A complete set of 100 seeds for a single cell takes  $\approx 30$  hours (parallelized to less than 5 hours on a 6-core machine).
- **EDP+ZI** ( $N = 500$ ): A single run takes approximately **27.5 hours**. This represents a  $\approx 90\times$  slowdown compared to a single run of ZIC-BCF-Smear.
- **EDP+ZI** ( $N = 1000$ ): Due to the  $O(N^3)$  complexity bottleneck, scaling the sample size from  $N = 500$  to  $N = 1000$  increases the runtime by a factor of

approximately  $(1000/500)^3 = 8$ . A single fit of EDP+ZI at  $N = 1000$  is estimated to take  $\approx 220$  hours (**more than 9 days**) of continuous single-threaded execution.

To run the full simulation grid for EDP+ZI (comprising 3 DGPs, 4 sample sizes, 4 zero-inflation levels, and 100 seeds) would require over 108,000 CPU hours (approximately 12 years of continuous runtime on a single machine). Given that standard academic cloud environments (like Google Colab) cap continuous sessions at 12 hours, running EDP+ZI across the full design was computationally infeasible. Consequently, while the model was verified for correctness, it was excluded from the empirical benchmark. This highlights that computational efficiency is a primary methodological advantage of ZIC-BCF-Smear.

## 4 Complete Simulation Results & Full Figures

The main manuscript summarizes both sensitivity sweeps in prose and reports no sensitivity figures, because the numerical divergence (overflow) of the parametric models under misspecification generates extreme outlier values (RMSE up to  $10^{11}$ ), which swamps the y-axis and makes visual comparison of the remaining stable models impossible. Both variants are therefore given here: the full 5-model figures including all benchmarks and diverged runs in this subsection and the next, and the legible variants restricted to the numerically stable estimators in Section 4.3.

### 4.1 Linear Sensitivity Plots

Under link-linear DGPs, the parametric Gamma Hurdle model is correctly specified. However, the naive Gamma + .01 model (which adds a constant of 0.01 to the zero outcomes) still suffers from severe bias and numerical instability.

- Figure 1 displays the full  $N$ -sensitivity plots for DGP A, B, and C. The naive Gamma + .01 model exhibits large RMSE values at small sample sizes ( $N = 100$ ) where the shift introduces massive bias relative to the small active set.
- Figure 2 displays the full zero-inflation sensitivity plots. As the zero proportion increases to 85%, the performance of the parametric Gamma Hurdle begins to degrade because the active sample size contracts, while Gamma + .01 collapses entirely.

### 4.2 Nonlinear Sensitivity Plots

Under link-nonlinearity, both parametric models (Gamma Hurdle and Gamma + .01) suffer from severe functional-form misspecification. The resulting out-of-sample prediction overflows lead to massive divergence in 4%–9% of the seeds, inflating the mean CATE and ATE RMSEs.

- Figure 3 displays the full  $N$ -sensitivity plots under nonlinearity. The y-axis scales are dominated by the diverged parametric model metrics, which reach values of  $10^8$  to  $10^{10}$ , showing the extreme risk of relying on parametric log-link models when the true surface is nonlinear.
- Figure 4 displays the full zero-inflation sensitivity plots under nonlinearity, illustrating that the numerical instability of the parametric benchmarks is present across all levels of zero-inflation.

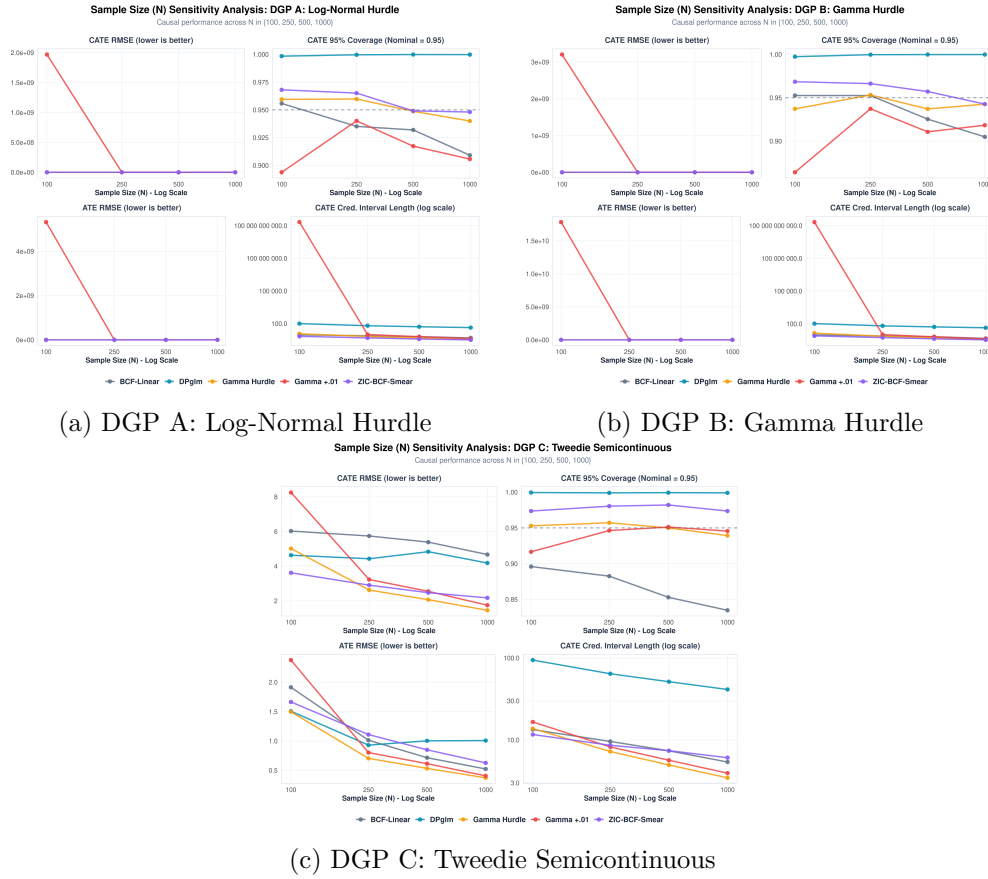


Figure 1: Full link-linear  $N$ -sensitivity analysis (including the diverged Gamma + .01 benchmark) across DGP A, B, and C.

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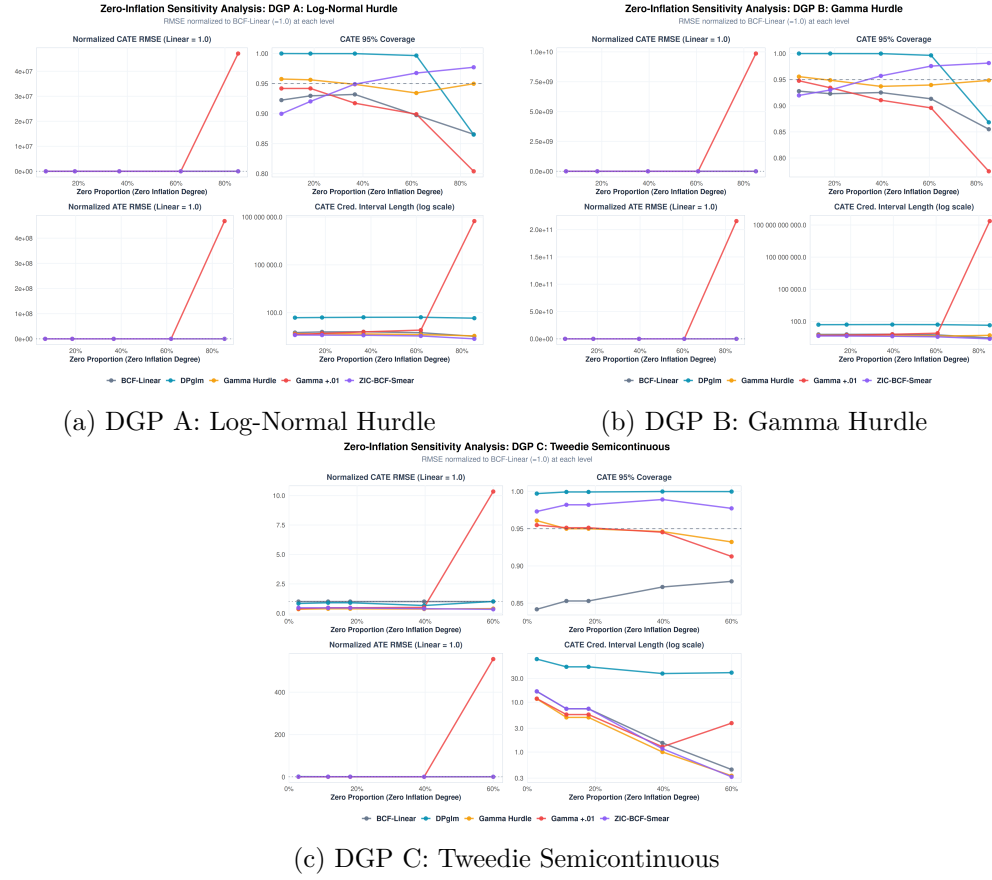


Figure 2: Full link-linear zero-inflation sensitivity analysis (including the diverged Gamma + .01 benchmark) across DGP A, B, and C.

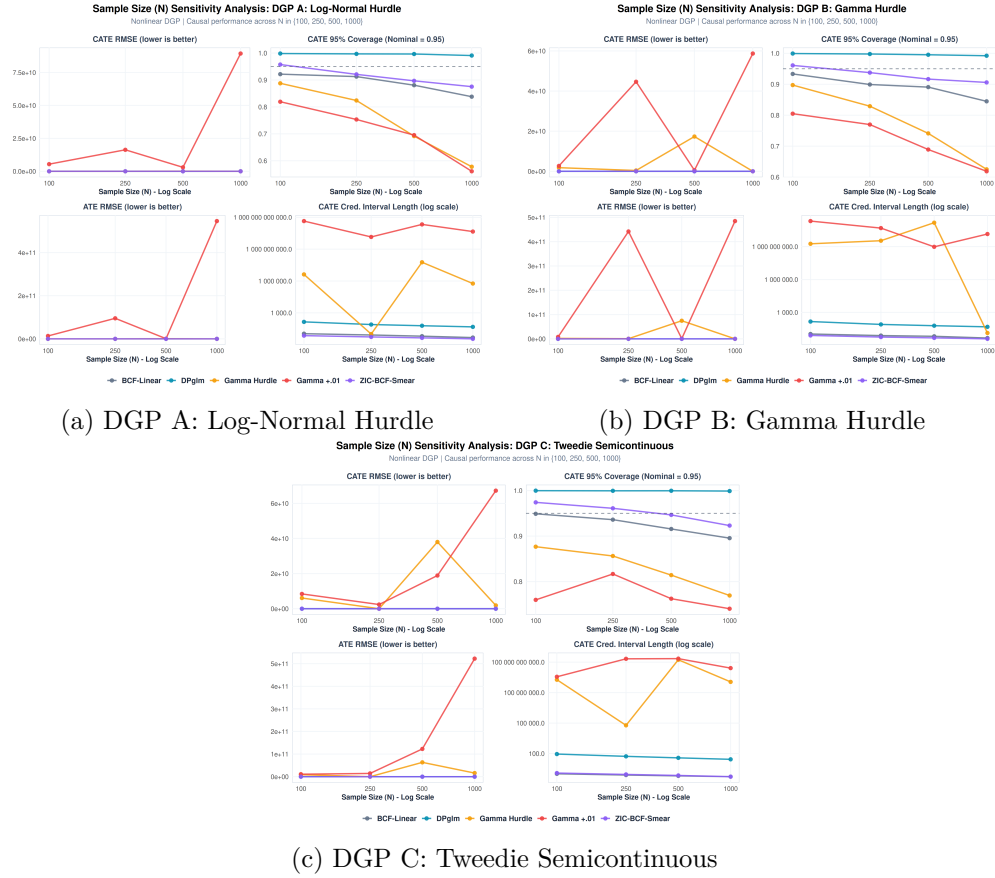


Figure 3: Full link-nonlinear  $N$ -sensitivity analysis under misspecification, displaying the extreme y-axis distortion caused by the diverged parametric benchmarks.

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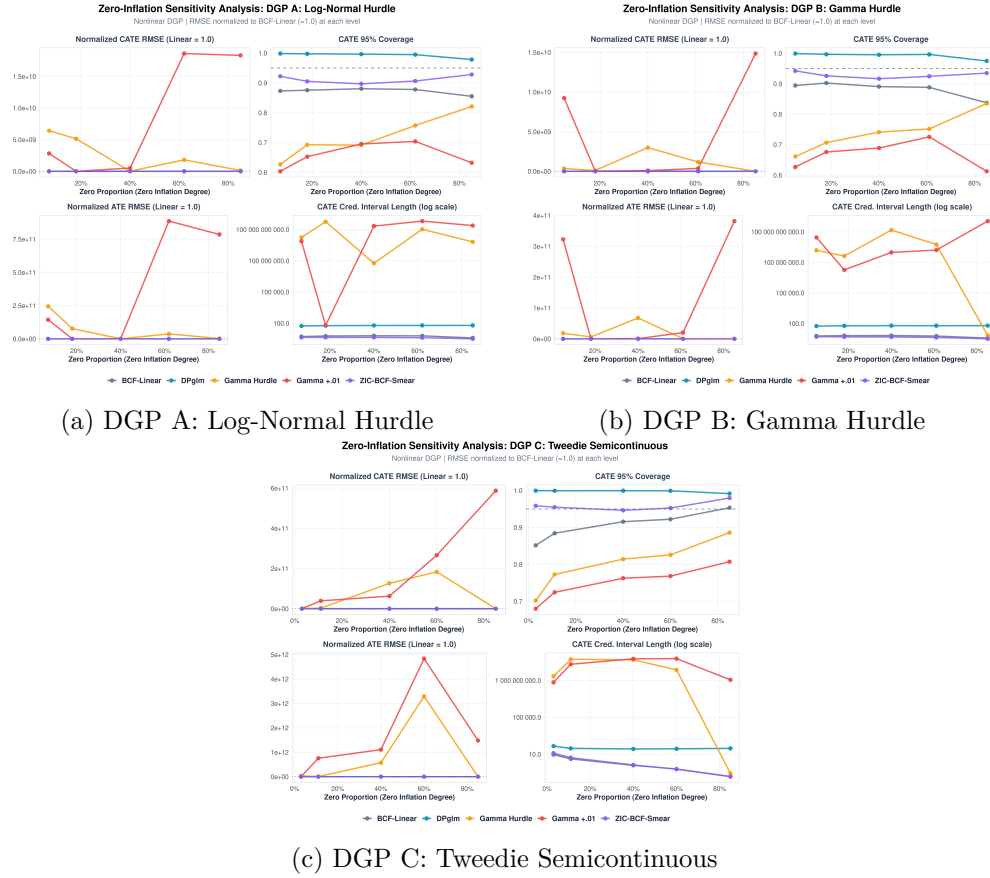


Figure 4: Full link-nonlinear zero-inflation sensitivity analysis under misspecification, displaying the extreme y-axis distortion caused by the diverged parametric benchmarks.

### 4.3 Legible Sensitivity Plots (Stable Estimators Only)

The main manuscript reports both sensitivity sweeps numerically rather than graphically, because the diverged parametric benchmarks destroy the y-axis scale of any plot that includes them. This subsection supplies the graphical counterpart with the diverged benchmarks removed, so that the numerically stable estimators can be compared on a readable scale. Figure 5 and Figure 6 give the link-linear  $N$ -sensitivity and zero-inflation-sensitivity analyses for DGP A with the diverged Gamma + .01 benchmark removed; Figure 7 and Figure 8 give the corresponding link-nonlinear analyses for all three DGPs with both diverged parametric benchmarks (Gamma Hurdle and Gamma + .01) removed.

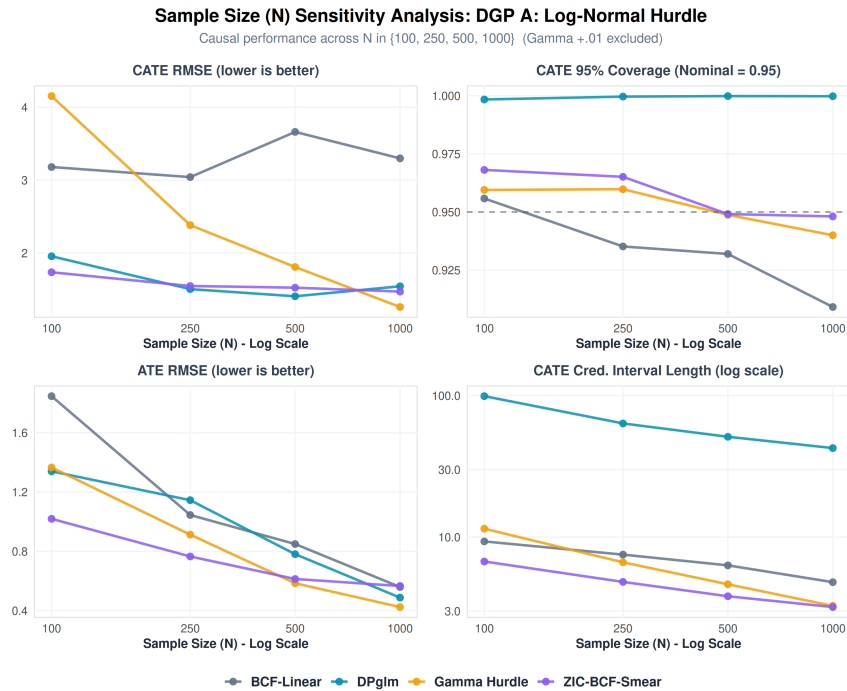


Figure 5: Link-linear  $N$ -sensitivity analysis for DGP A (Log-Normal Hurdle), excluding the diverged Gamma +.01 benchmark for legibility. Reading the four panels together, ZIC-BCF-Smear and the correctly specified Gamma Hurdle both converge as  $N$  grows, but ZIC-BCF-Smear reaches the lower CATE RMSE with the shorter credible intervals (log scale, bottom right) while holding coverage at the nominal 95%; DPglm's intervals remain an order of magnitude wider at every sample size, which is why its coverage is pinned at 100%.

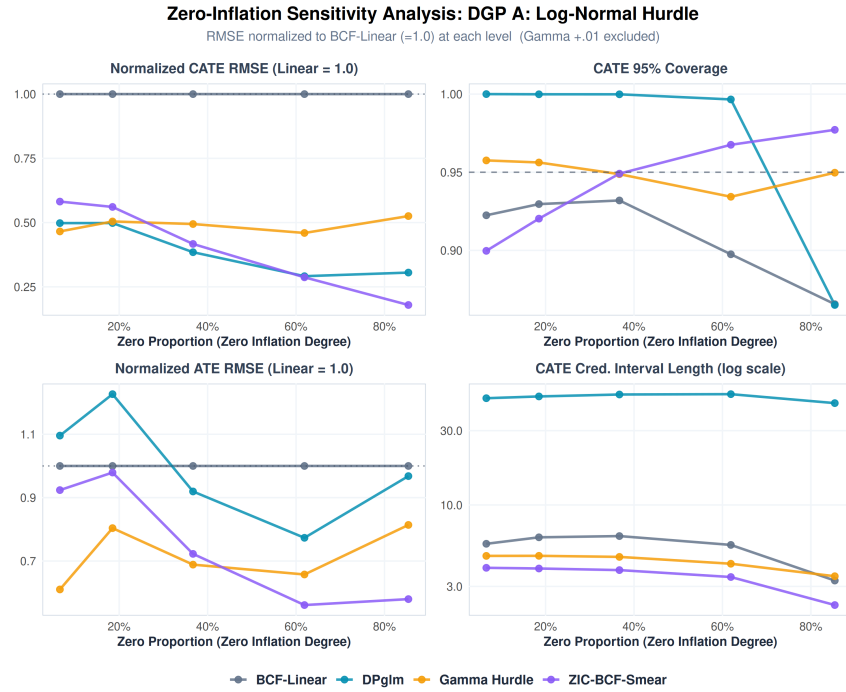


Figure 6: Link-linear zero-inflation sensitivity analysis for DGP A (Log-Normal Hurdle), excluding the diverged Gamma +.01 benchmark for legibility. RMSE is normalized by the true CATE scale so that cells with different zero proportions are comparable. The ZIC-BCF-Smear advantage over the single-part BCF-Linear baseline widens monotonically as the zero proportion rises toward 85%, while the correctly specified Gamma Hurdle deteriorates at the right-hand end of the range as the active sample shrinks below  $\approx 15\%$  of the data.

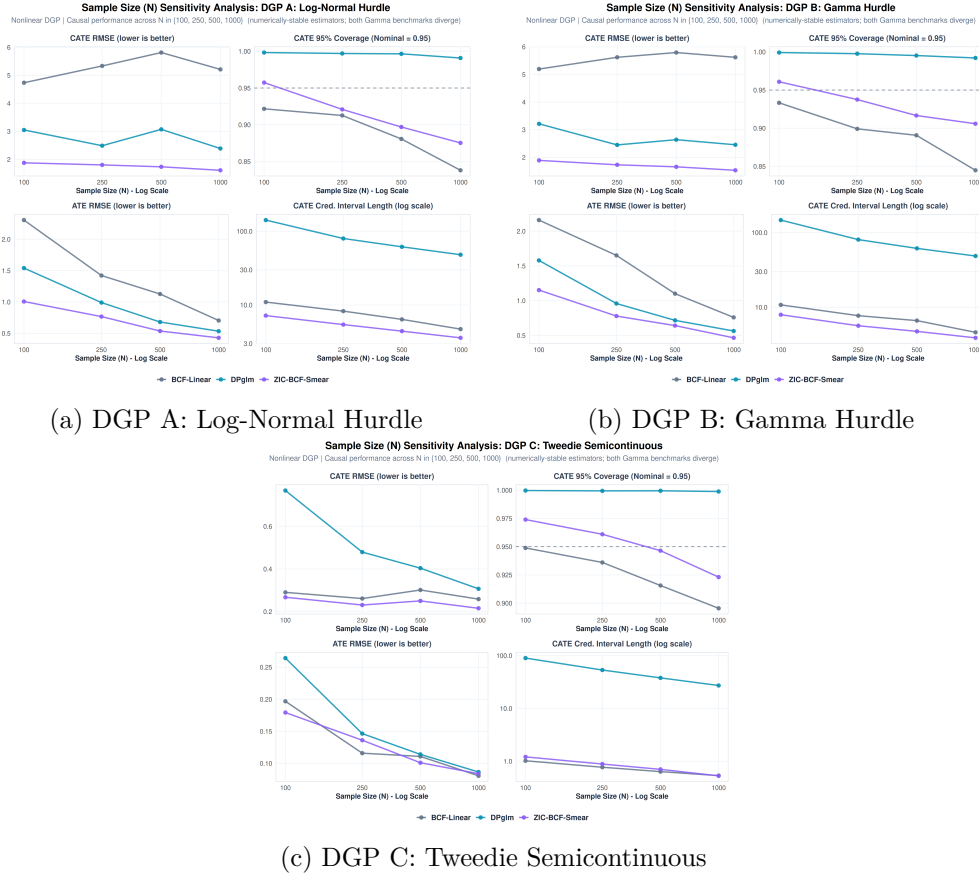


Figure 7: Link-nonlinear  $N$ -sensitivity analysis restricted to the numerically stable estimators (BCF-Linear, DPglm, and ZIC-BCF-Smear). Reading across the panels, ZIC-BCF-Smear holds the lowest CATE RMSE and the coverage closest to the nominal 95% at every sample size in all three DGPs, while DPglm's credible-interval length stays an order of magnitude wider throughout.

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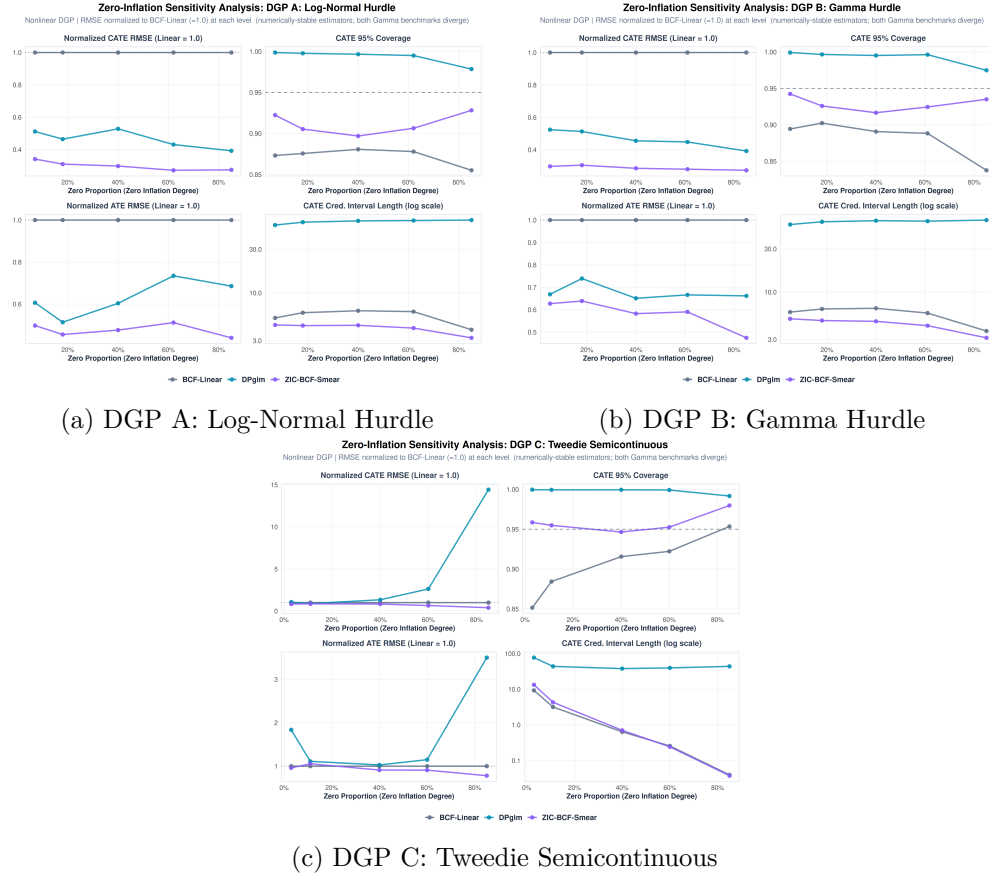


Figure 8: Link-nonlinear zero-inflation sensitivity analysis restricted to the numerically stable estimators. The gap between ZIC-BCF-Smear and the single-part BCF-Linear baseline widens monotonically as the zero proportion rises toward 85%, which is the graphical signature of fitting the intensity forest only on units that clear the hurdle.

#### 4.4 Numerical Stability of the Parametric Benchmarks

Table 3 reports the divergence rates of the parametric benchmarks under the link-nonlinear DGPs, together with medians computed across the surviving seeds. The medians are given so that the comparison does not rest solely on means contaminated by overflow. They show that excluding the diverged seeds does not rescue the benchmarks: the CATE correlations remain negative, and the robust median CATE RMSEs remain a factor of 2.0 to 2.7 above ZIC-BCF-Smear.

Table 3: Numerical stability of the parametric benchmarks under nonlinear DGPs ( $N = 500$ , ZI Level 3). Reports divergence rates and robust medians across 100 seeds.

| DGP               | Model                | Diverged Seeds | CATE RMSE (median) | ATE Abs Err (median) | CATE Corr    |
|-------------------|----------------------|----------------|--------------------|----------------------|--------------|
| DGP A: Log-Normal | Gamma Hurdle         | 4 / 100        | 3.960              | 0.759                | -0.039       |
|                   | Gamma +.01           | 5 / 100        | 4.336              | 0.747                | -0.081       |
|                   | <b>ZIC-BCF-Smear</b> | <b>0 / 100</b> | <b>1.611</b>       | <b>0.352</b>         | <b>0.686</b> |
| DGP B: Gamma      | Gamma Hurdle         | 5 / 100        | 3.704              | 0.784                | -0.040       |
|                   | Gamma +.01           | 9 / 100        | 4.145              | 0.692                | -0.068       |
|                   | <b>ZIC-BCF-Smear</b> | <b>0 / 100</b> | <b>1.537</b>       | <b>0.494</b>         | <b>0.645</b> |
| DGP C: Tweedie    | Gamma Hurdle         | 5 / 100        | 0.369              | 0.122                | -0.060       |
|                   | Gamma +.01           | 7 / 100        | 0.460              | 0.147                | -0.045       |
|                   | <b>ZIC-BCF-Smear</b> | <b>0 / 100</b> | <b>0.188</b>       | <b>0.065</b>         | <b>0.175</b> |

#### 4.5 Hurdle Stage Sensitivity Results

For completeness, the full numerical values of the Hurdle stage sensitivity analyses are provided. The standard configuration ( $N = 500$ ) appears as the middle column of each table, and reproduces the hurdle-stage figures quoted in the main manuscript. Table 4 displays the Hurdle stage metrics across sample size  $N$  and zero-inflation levels for the link-linear study. Table 5 displays the Hurdle stage metrics for the link-nonlinear study.

Table 4: Linear study Hurdle stage sensitivity results across sample sizes  $N$  (under nominal zero-inflation) and zero-inflation levels (under  $N = 500$ ).

| DGP               | Model                | Parameter | $N = 100$    | $N = 250$    | $N = 500$    | $N = 1000$   | ZI 85% (Level 1) |
|-------------------|----------------------|-----------|--------------|--------------|--------------|--------------|------------------|
| DGP A: Log-Normal | DPglm                | RMSE      | 0.086        | 0.067        | 0.053        | 0.057        | 0.050            |
|                   |                      | Coverage  | 98.0%        | 99.5%        | 99.7%        | 99.7%        | 86.6%            |
|                   | Gamma Hurdle         | RMSE      | 0.208        | 0.129        | 0.098        | 0.071        | 0.080            |
|                   |                      | Coverage  | 95.1%        | 95.8%        | 94.0%        | 94.8%        | 95.1%            |
|                   | <b>ZIC-BCF-Smear</b> | RMSE      | <b>0.068</b> | <b>0.054</b> | <b>0.046</b> | <b>0.040</b> | <b>0.036</b>     |
|                   |                      | Coverage  | <b>98.9%</b> | <b>99.0%</b> | <b>99.3%</b> | <b>98.6%</b> | <b>98.3%</b>     |
| DGP B: Gamma      | DPglm                | RMSE      | 0.096        | 0.065        | 0.055        | 0.056        | 0.047            |
|                   |                      | Coverage  | 97.9%        | 99.5%        | 99.7%        | 99.6%        | 86.9%            |
|                   | Gamma Hurdle         | RMSE      | 0.208        | 0.129        | 0.098        | 0.071        | 0.081            |
|                   |                      | Coverage  | 95.0%        | 95.8%        | 94.0%        | 94.8%        | 95.0%            |
|                   | <b>ZIC-BCF-Smear</b> | RMSE      | <b>0.067</b> | <b>0.054</b> | <b>0.046</b> | <b>0.040</b> | <b>0.036</b>     |
|                   |                      | Coverage  | <b>98.5%</b> | <b>98.8%</b> | <b>99.3%</b> | <b>98.3%</b> | <b>98.6%</b>     |
| DGP C: Tweedie    | DPglm                | RMSE      | 0.070        | 0.052        | 0.044        | 0.036        | 0.059            |
|                   |                      | Coverage  | 81.3%        | 84.3%        | 87.0%        | 88.3%        | 100.0%           |
|                   | Gamma Hurdle         | RMSE      | 0.173        | 0.114        | 0.081        | 0.061        | 0.097            |
|                   |                      | Coverage  | 97.5%        | 96.8%        | 95.0%        | 92.7%        | 94.4%            |
|                   | <b>ZIC-BCF-Smear</b> | RMSE      | <b>0.044</b> | <b>0.035</b> | <b>0.028</b> | <b>0.026</b> | <b>0.045</b>     |
|                   |                      | Coverage  | <b>95.3%</b> | <b>98.2%</b> | <b>99.3%</b> | <b>97.9%</b> | <b>97.9%</b>     |

Table 5: Nonlinear study Hurdle stage sensitivity results across sample sizes  $N$  (under nominal zero-inflation) and zero-inflation levels (under  $N = 500$ ).

| DGP               | Model                | Parameter | $N = 100$    | $N = 250$    | $N = 500$    | $N = 1000$   | ZI 85% (Level 1) |
|-------------------|----------------------|-----------|--------------|--------------|--------------|--------------|------------------|
| DGP A: Log-Normal | DPglm                | RMSE      | 0.105        | 0.085        | 0.077        | 0.078        | 0.078            |
|                   |                      | Coverage  | 98.2%        | 98.8%        | 98.6%        | 96.1%        | 96.9%            |
|                   | Gamma Hurdle         | RMSE      | 0.229        | 0.167        | 0.129        | 0.111        | 0.104            |
|                   |                      | Coverage  | 92.6%        | 89.9%        | 88.6%        | 82.2%        | 85.9%            |
|                   | <b>ZIC-BCF-Smear</b> | RMSE      | <b>0.083</b> | <b>0.074</b> | <b>0.065</b> | <b>0.062</b> | <b>0.065</b>     |
|                   |                      | Coverage  | <b>98.0%</b> | <b>96.4%</b> | <b>94.7%</b> | <b>90.8%</b> | <b>92.8%</b>     |
| DGP B: Gamma      | DPglm                | RMSE      | 0.101        | 0.088        | 0.077        | 0.077        | 0.080            |
|                   |                      | Coverage  | 98.4%        | 98.8%        | 98.2%        | 96.6%        | 96.7%            |
|                   | Gamma Hurdle         | RMSE      | 0.229        | 0.166        | 0.129        | 0.111        | 0.104            |
|                   |                      | Coverage  | 92.6%        | 90.0%        | 88.5%        | 82.3%        | 86.0%            |
|                   | <b>ZIC-BCF-Smear</b> | RMSE      | <b>0.083</b> | <b>0.075</b> | <b>0.066</b> | <b>0.062</b> | <b>0.065</b>     |
|                   |                      | Coverage  | <b>98.0%</b> | <b>96.4%</b> | <b>94.5%</b> | <b>90.9%</b> | <b>92.9%</b>     |
| DGP C: Tweedie    | DPglm                | RMSE      | 0.099        | 0.067        | 0.057        | 0.049        | 0.051            |
|                   |                      | Coverage  | 99.6%        | 99.9%        | 99.8%        | 99.6%        | 98.3%            |
|                   | Gamma Hurdle         | RMSE      | 0.220        | 0.150        | 0.111        | 0.090        | 0.083            |
|                   |                      | Coverage  | 93.7%        | 93.5%        | 92.4%        | 88.1%        | 93.6%            |
|                   | <b>ZIC-BCF-Smear</b> | RMSE      | <b>0.069</b> | <b>0.051</b> | <b>0.044</b> | <b>0.040</b> | <b>0.028</b>     |
|                   |                      | Coverage  | <b>99.2%</b> | <b>99.7%</b> | <b>98.4%</b> | <b>96.6%</b> | <b>98.5%</b>     |

## 5 Applied Study: Severe Adverse-Event Burden in the SPECTRUM Head-and-Neck Cancer Trial

This section is the extended, self-contained version of the oncology applied study summarized in the main manuscript. It develops in full the secondary causal analysis of the SPECTRUM phase III trial ([Vermorken et al., 2013](#); [ClinicalTrials.gov, 2026](#)), whose individual-participant data were obtained through Project Data Sphere ([Project Data Sphere, 2026](#)): the analytic cohort, the two outcome definitions and the estimand they identify, the posterior models and their computation, the complete subgroup and contrast results under both endpoints, the reconciliation with the clinical and statistical literature, the limitations, and the reproducibility trail.

### 5.1 Trial Context and Analytic Cohort

SPECTRUM was an open-label phase III randomized trial of cisplatin and fluorouracil with or without panitumumab for recurrent or metastatic squamous-cell carcinoma of the head and neck ([Vermorken et al., 2013](#); [ClinicalTrials.gov, 2026](#)). The parent trial reported 657 randomized patients. The current analytic extract contains 520 participants, balanced at 260 per arm.

#### What the Cohort Derivation Can and Cannot Establish

The step from 657 to 520 deserves to be stated precisely, because randomization is what licenses the causal reading and randomization attaches to the 657, not automatically to an analytically constructed subset.

What is checkable from the supplied files is the following. The analytic cohort is the set of subjects in the supplied core-variable file with a codeable treatment arm, one row per subject, and it contains exactly 260 patients per arm. The adverse-event file contains 9,934 records on 514 distinct subjects, 256 in the chemotherapy arm and 258 in the panitumumab arm; the 6 analytic-cohort patients with no adverse-event record contribute an outcome of zero at every horizon. Recomputing the summed-duration endpoint from the adverse-event file reproduces the patient-level outcome for all 520 patients exactly, so the linkage between the two files is sound. That reconciliation, from 520 to 514 and back, is complete.

What is not checkable is the step from 657 to 520. The reduction occurs in the supplied extract itself rather than in any analysis code: the core-variable file distributed with the `HeadNe_Amgen_2007_265` package contains those 520 subjects, and no file in the package records which parent-trial patients were omitted or why. A reader therefore cannot tell from these materials whether the reduction is a benign completeness filter or a selection that differs by arm, and the exactly balanced 260 per arm is consistent with either a deliberately balanced release or a coincidence. This bears directly on the intention-to-treat characterization: the estimand below is the average treatment effect in the 520-person analytic cohort, and the extent to which that transfers to the parent trial’s randomized population is not established here. Resolving it requires the release

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documentation for the extract, which would have to be requested from Project Data Sphere.

### Treatment Coding and the Adjustment Set

Treatment is coded as  $Z_i = 1$  for panitumumab plus chemotherapy and  $Z_i = 0$  for chemotherapy alone. The baseline covariate vector  $X_i$  contains age, ECOG status, sex, diagnosis site, HPV status, disease status, and the patient-level prior-radiotherapy indicator PRRADIO. Because the trial randomized assignment, the design propensity is known to be 0.5 and is supplied directly to ZIC-BCF-Smear rather than estimated from the realized arms. The unadjusted arm difference is therefore already unbiased for the cohort average effect.

Two changes to the adjustment set relative to the first version of this study should be recorded, together with their justifications.

Tumor category is removed. In this extract TUMCAT is a deterministic recode of DIAGTYPE: larynx and oropharynx map to Y, hypopharynx and oral cavity map to N, with no exceptions among the 520 patients. Including both therefore places an exact linear combination of existing columns in the design matrix. The recode is verified rather than asserted, by `prepare_analysis_data.R`, which halts if any diagnosis site maps to more than one tumor category. Tumor category is also dropped from the prespecified subgroup list, since it is a coarsening of a grouping that is already reported at full resolution.

Prior treatment for squamous-cell carcinoma of the head and neck (PRHNTRTC) remains in the released patient-level analysis data as a descriptive variable, while the model adjustment set and the prespecified subgroup lists use the now-available prior-radiotherapy indicator PRRADIO. PRHNTRTC is present for 204 of 260 chemotherapy patients (78.5%) and 216 of 260 panitumumab patients (83.1%); PRRADIO is present for 200 chemotherapy patients (76.9%) and 216 panitumumab patients (83.1%). Prior radiotherapy is a principal clinical candidate for mucosal and cutaneous toxicity, the outcome being modeled. The two indicators differ for only four patients, all in the chemotherapy arm, so they are not entered together: their near-redundancy destabilized the Gamma benchmark, whereas PRRADIO provides the clinically specific adjustment and subgroup variable.

Three qualifications attach to these variables. PRHNTRTC is a composite: the data definition table derives it as Y if the subject had prior radiotherapy as per RAHX, or other prior treatment as per TXHX or OTHEPROC ([Amgen Inc., 2013](#)). The v2 demo file supplies PRRADIO, and the radiotx file supplies 961 detailed radiotherapy records covering all 520 subjects, 857 of them with RAANY = Y at the record level and every patient recorded as PRRADIO = N carrying an explicit RAANY = N record. The subject-level agreement between RAANY and PRRADIO for all 520 patients (416 versus 104) is therefore a genuine coding concordance rather than an artifact of absent files being coalesced to no radiotherapy. The lesion file supplies LSPIRRNY, and that flag does not merely leave gaps relative to PRRADIO but contradicts it often enough to be informative: 42 of the 104 patients recorded as PRRADIO = N (40.4%) have at least

one lesion marked previously irradiated, against 110 of the 416 patients recorded as  $\text{PRRADIO} = Y$  (26.4%), and among the 373 patients with any lesion-level information the corresponding rates are 50.0% and 38.1%. A flag more often positive among patients recorded as having had no prior radiotherapy is either a different construct, for instance a lesion treated in another field or for another indication, or a merge or level-coding issue; the flag is therefore not used as a covariate and this cross-tabulation is reported rather than passed over in silence. The extract also carries  $\text{PRHNTRTI}$ , derived from the randomization stratification variable, and the two prior-treatment versions agree for 501 of 520 patients (96.3%). Finally, the treated and untreated strata carry the same exposure imbalance as the cohort as a whole, so a difference between them cannot be attributed to prior treatment or prior radiotherapy alone.

## 5.2 Outcome Construction and Estimand

Let  $\mathcal{A}_i$  denote the on-study AE records for patient  $i$ , let  $d_{ij}$  be the recorded duration of event  $j$ , and let  $G_{ij}$  be its recorded grade. The initial patient-level outcome is

$$Y_i^{\text{sum}} = \sum_{j \in \mathcal{A}_i} d_{ij} \mathbf{1}(G_{ij} \geq 3), \quad (5.1)$$

where records explicitly marked as pre-treatment are excluded and severe records without a recorded duration contribute zero in the current implementation. Equation (5.1) is a sum of event-record durations. It is not the union of severe-AE calendar-date intervals. Accordingly, simultaneous severe events can add more than one event-day on the same date, while recurrence adds further event-days. This outcome should be called *summed severe-AE-record duration* or *severe-AE event-day burden*, rather than distinct days in a severe-AE state.

Of 9,934 AE records, 8,518 have a recorded duration. Among 1,718 grade 3+ records, 1,589 have a recorded duration (92.5%). The missing-duration records are not evidence that the corresponding severe event had zero duration. They remain an outcome-reconstruction limitation. The summed-duration outcome has 123 zero values (23.7%), 397 positive values, a median positive value of 26, and a maximum of 641 event-days. Such zero-inflated nonnegative continuous outcomes require both the point mass and the skewed positive distribution to be handled explicitly (Liu et al., 2019).

### Distinct Severe-AE Calendar Days

The companion outcome counts the distinct on-study calendar days on which patient  $i$  experienced at least one grade 3+ AE. Let  $s_{ij}$  and  $e_{ij}$  denote the imputed start and end study days for a qualifying record with complete interval boundaries, and let  $\mathcal{C}_i$  be the set of these records. The endpoint is

$$Y_i^{\text{cal}} = \left| \bigcup_{j \in \mathcal{C}_i} \{s_{ij}, s_{ij} + 1, \dots, e_{ij}\} \right|. \quad (5.2)$$

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The interval is inclusive, consistently with the data dictionary definition of AE duration as the end date minus the start date plus one. Within each patient, overlapping or adjacent intervals are merged computationally and each calendar day is counted once. Equation (5.2) is an outcome-definition sensitivity analysis, not a correction to Equation (5.1): summed duration gives additional weight to concurrent severe conditions, whereas distinct days measure observed time with any severe toxicity.

The original subject identifier in the 9,934-record AE file links exactly to the patient-level file containing the 520 randomized patients, of whom 514 have at least one AE record. Recomputing Equation (5.1) from the AE file reproduces the existing patient-level outcome for all 520 patients. The distinct-day endpoint uses records not explicitly identified as prior to treatment and with severity grade at least 3.

### Accounting for Unresolved Interval Boundaries

Of 1,671 qualifying records, 1,547 (92.6%) have complete imputed start and end study days. The remaining 124 records have an unresolved boundary. The v2 disposition file now supplies DSDY and LASTCTDY for every patient, but those variables are observation or contact boundaries, not missing event end dates, and only 354 patients completed the safety follow-up. Complete intervals reproduce the corresponding recorded durations exactly. Thus, Equation (5.2) is more directly interpretable than summed record duration but remains an observed-complete-interval endpoint.

Two consequences of that construction were previously left for a reader to infer, and are reported here instead. The endpoint builder retains only records with a complete imputed interval and assigns zero to any patient with none, so both counts matter. In all 124 unresolved records the missing boundary is the end day rather than the start day, without exception. Those records fall across 88 patients, 51 on panitumumab and 37 on chemotherapy, whose distinct-day totals therefore omit some severe-adverse-event days. A smaller group of 13 patients has no complete qualifying interval at all and is assigned zero outright, and that group is imbalanced 10 on panitumumab against 3 on chemotherapy.

This should not be over-read. The adverse-event file carries no ongoing-at-cutoff flag, so a missing end day cannot be shown to mean the event was still running when data collection stopped. The new disposition fields show that 93 complete intervals end after DSDY but only 3 end after LASTCTDY, supporting LASTCTDY as a useful observed upper bound for a diagnostic, not as proof of complete safety ascertainment. Six unresolved intervals also begin after DSDY itself, which is one further reason DSDY cannot serve directly as a censoring variable while LASTCTDY remains the better observed window. What is measurable is the 10-versus-3 imbalance among the forced zeros. It runs in the direction of understating the panitumumab arm, and because it acts on the zero indicator it lands specifically on the hurdle component, which is the component this study reports as uncertain. The imbalance is small relative to the 520-patient cohort but it is not neutral with respect to the conclusion it touches.

### The Estimand

For either outcome  $k \in \{\text{sum}, \text{cal}\}$ , the causal estimand is the analytic-cohort average treatment effect

$$\Delta^k = \mathbb{E}\{Y_i^k(1) - Y_i^k(0) \mid i \in \mathcal{S}\}, \quad (5.3)$$

where  $\mathcal{S}$  is the 520-person analytic cohort. Under random assignment, consistency, and adequate outcome reconstruction, Equation (5.3) is the effect of *assignment* on accrued event-day burden and carries an intention-to-treat interpretation within  $\mathcal{S}$ , subject to the caveat of Section 5.1 about how  $\mathcal{S}$  was formed.

Three things Equation (5.3) is not should be stated plainly, because the third is the one most easily mistaken. It is not a fixed-horizon restricted mean. It is not a competing-risk estimand: death, progression, and discontinuation are post-randomization events that affect accumulated burden and are not modeled. And it is not a rate per unit of treatment exposure. Treatment duration differs sharply between the arms, and structurally so, because panitumumab continues as maintenance after chemotherapy stops. That difference is a consequence of assignment, so it lies on the causal pathway from assignment to accrued burden: it is part of the effect being estimated rather than a nuisance to be removed, and conditioning on it would both block part of the effect and condition on a post-treatment variable. Equation (5.3) is therefore the effect of assignment under the trial's own differential treatment duration, and Section 5.3 quantifies how much of the effect travels through that channel.

Table 6: Observed summed-duration outcome by randomized arm. Positive-part summaries condition on  $Y_i^{\text{sum}} > 0$ .

| Arm                        | $n$ | Zero (%) | Mean  | Positive mean | Positive median | Maximum |
|----------------------------|-----|----------|-------|---------------|-----------------|---------|
| Chemotherapy               | 260 | 24.6     | 26.75 | 35.49         | 20.5            | 263     |
| Panitumumab + chemotherapy | 260 | 22.7     | 42.43 | 54.89         | 36.0            | 641     |

Table 7: Descriptive comparison of the summed-duration and distinct-calendar-day endpoints. The unadjusted difference is panitumumab plus chemotherapy minus chemotherapy.

| Endpoint                         | Total days | Chemotherapy mean | Panitumumab + chemotherapy mean | Unadjusted difference | Zeros |
|----------------------------------|------------|-------------------|---------------------------------|-----------------------|-------|
| Summed AE-record days            | 17,988     | 26.75             | 42.43                           | 15.68                 | 123   |
| Distinct severe-AE calendar days | 13,107     | 19.18             | 31.23                           | 12.06                 | 123   |

The interval union removes 4,881 record-days (27.1% of the original total) across 217 patients. It reduces the unadjusted arm difference by 3.62 days (23.1%) without changing the zero-outcome count. Every positive recorded duration in this extract has an associated complete date interval, which explains why the hurdle indicator is unchanged.

### 5.3 Treatment Exposure, Common-Horizon Truncation, and Person-Time

The endpoints of Section 5.2 count days on which a patient was in a severe-adverse-event state. That quantity accrues with time on treatment, so it is interpretable as a between-

arm comparison only if the reader is told how far the arms differ in exposure. They differ a great deal, and the difference is structural rather than incidental: panitumumab continues as maintenance after chemotherapy stops. Table 8 reports the distribution of TRTDUR, defined in the data definition table as the last dose date minus the first dose date plus one (Amgen Inc., 2013), so it measures time on treatment and not length of follow-up.

Table 8: Treatment duration (TRTDUR, days) by randomized arm in the 520-person analytic cohort. TRTDUR is a post-randomization quantity and appears in no adjustment set.

| Arm                           | <i>n</i> | Mean   | SD     | Median | Maximum |
|-------------------------------|----------|--------|--------|--------|---------|
| Chemotherapy                  | 260      | 83.72  | 50.14  | 89.0   | 276     |
| Panitumumab plus chemotherapy | 260      | 130.30 | 127.19 | 112.5  | 1,149   |

Mean time on treatment is 56% longer under panitumumab, and the maxima differ by a factor of four. The consequence for the endpoint is large. Table 9 truncates every patient’s observed accrual at a common study day and recomputes the unadjusted arm difference in distinct severe-adverse-event days. The v2 disposition file adds a separate observed-window diagnostic based on LASTCTDY, reported below, but it does not turn treatment duration into a follow-up measure.

Two cautions govern how Table 9 is read, and they matter more than the numbers in it.

The first is that this is truncation of observed accrual, not a fixed-horizon estimand. Capping at day 84 removes events recorded after day 84, but it does not create observation for a patient whose follow-up ended before day 84. The v2 disposition file supplies LASTCTDY for all 520 patients, and 47 chemotherapy patients and 41 panitumumab patients have last contact before day 84. Capping complete intervals at the patient-specific last-contact day gives mean observed-window burdens of 19.18 and 30.90 days by arm; at day 84, the corresponding means are 11.25 and 14.33 days, with 213 and 219 patients reaching that contact horizon. The horizon diagnostic checks the first alternative explanation of that collapse, that differential follow-up rather than exposure produces the attenuation, but it is not a corrected estimate. At day 84 the mean among patients whose contact actually reaches the horizon is 11.25 in the chemotherapy arm against 14.93 under panitumumab, a difference of 3.68 days that exceeds the 3.08-day difference over all patients; at day 120 the corresponding difference is 5.13 days (16.12 against 21.25) against 4.03 over all patients. The difference is therefore not smaller among patients observed through the horizon, so simple truncation of the shortest-observed records is not what produces the attenuation. Restricting to patients whose contact reaches the horizon conditions on a post-randomization variable, however, so informative censoring remains possible and the direction of any resulting selection bias cannot be signed in advance. These quantities remain diagnostics, not corrected fixed-horizon estimates, because early contact loss, incomplete event intervals, and competing events require an explicit censoring and state definition. Capping applies only to intervals with complete boundaries, so these diagnostics inherit, in bounded form,

Table 9: Mean distinct severe-adverse-event calendar days by arm under common-horizon truncation, on the 520-person basis with 260 patients per arm. The six patients with no adverse-event record contribute zero at every horizon. The primary row uses the imputed start and end days; the three remaining date variants recorded in the file are shown for comparison. This table is a diagnostic, not a corrected estimate; see the text.

| Date variant and horizon                    | Chemotherapy | Panitumumab | Difference |
|---------------------------------------------|--------------|-------------|------------|
| <i>Imputed start, imputed end (primary)</i> |              |             |            |
| Uncapped                                    | 19.18        | 31.23       | +12.06     |
| 120 days                                    | 15.55        | 19.65       | +4.10      |
| 90 days                                     | 11.96        | 15.33       | +3.37      |
| 84 days                                     | 11.25        | 14.39       | +3.14      |
| <i>Raw start, raw end</i>                   |              |             |            |
| Uncapped                                    | 18.83        | 29.27       | +10.44     |
| 84 days                                     | 10.90        | 13.35       | +2.44      |
| <i>Raw start, imputed end</i>               |              |             |            |
| Uncapped                                    | 19.06        | 30.18       | +11.12     |
| 84 days                                     | 11.13        | 13.77       | +2.64      |
| <i>Imputed start, raw end</i>               |              |             |            |
| Uncapped                                    | 18.95        | 29.85       | +10.90     |
| 84 days                                     | 11.02        | 13.64       | +2.62      |

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the forced-zero problem documented above. The attenuation in Table 9 also reproduces across all four date variants, so it is a property of the data rather than of one imputation convention.

The second is that whether the effect reverses on a per-exposure basis depends on the normalization, and the two natural normalizations disagree. Table 10 reports both.

Table 10: Descriptive per-exposure burden by arm. The patient-level mean weights every patient equally and therefore up-weights patients with short exposure; the aggregate rate pools days over pooled treatment-time. Neither is an adjusted causal effect.

| Arm                           | Total treatment days | Total severe-AE days | Patient-level mean rate | Aggregate rate |
|-------------------------------|----------------------|----------------------|-------------------------|----------------|
| Chemotherapy                  | 21,766               | 4,986                | 53.72                   | 22.91          |
| Panitumumab plus chemotherapy | 33,877               | 8,121                | 49.91                   | 23.97          |

The patient-level mean of per-100-treatment-day rates reverses the sign of the arm comparison, at 53.72 against 49.91, while the aggregate person-time rate does not, at 22.91 against 23.97. Fitting the same two models to the per-exposure rate reproduces that ambiguity rather than resolving it: ZIC-BCF-Smear estimates 5.06 severe-adverse-event days per 100 treatment-days (95% credible interval  $[-4.50, 17.40]$ ,  $P(> 0) = 0.837$ ) and the Gamma hurdle  $-5.23$  (95% credible interval  $[-37.06, 25.19]$ ,  $P(> 0) = 0.369$ ). Both are compatible with no effect. The Gamma benchmark’s exposure-rate estimate is also where its fragility appears across runs rather than within one: the preceding run, which carried both prior-treatment indicators, returned  $-3.47$  (95% credible interval  $[-33.40, 24.51]$ ) where the present run returns  $-5.23$ , a swing produced by nothing more than how a covariate differing for four patients was parameterized. The argument elsewhere that the parametric benchmark is fragile rests on this kind of behavior, so the movement between two defensible parameterizations of the same adjustment set belongs here rather than in a code comment. The defensible summary is severe attenuation once exposure is accounted for, not reversal.

#### Why This Is Reported as Descriptive

Everything in this subsection is descriptive, and the randomized analysis of Section 5.4 is left standing rather than replaced. The reason is not caution for its own sake. Treatment duration is measured after randomization, so dividing by it conditions on a post-treatment variable and can induce selection bias. The direction of that bias is not fixed in advance and is not recoverable from these files. A patient who discontinues early because of toxicity contributes a short denominator alongside a large numerator, which inflates that patient’s rate and makes their arm look worse; a patient who discontinues early for an unrelated reason pushes the same arm’s rate down. Which of the two dominates depends on why patients discontinued, and the extract records no discontinuation reason. The per-exposure quantities are therefore labeled descriptive rates per treatment-day rather than adjusted causal effects, and the truncation table is labeled a diagnostic.

Three separate things run together in this material and are worth keeping apart. Treatment exposure is measured, and is reported here. Patient-level contact and safety-follow-up boundaries are now also measured, but safety follow-up was completed by only

354 patients, so they support censoring diagnostics rather than a fully observed fixed-horizon safety endpoint. And the estimand itself is affected by treatment duration being a post-randomization consequence of assignment, so duration can be neither ignored in interpretation nor naively conditioned on in estimation. The v2 domains therefore make a fixed-horizon analysis feasible in principle, but not assumption-free or automatically identified from the observed endpoint.

The right target remains a prespecified fixed-horizon restricted-mean or multistate formulation. The newly available domains remove the need to request a basic last-contact field, but a valid analysis would still require an explicit censoring rule, a treatment policy, and a decision about death, progression, and discontinuation. If instead a causal quantity that removes the duration channel is wanted from the existing data, it requires an explicit assumption about a treatment-duration policy, stated as such. An offset on treatment duration is not that assumption and should not be presented as one.

Table 11: Disposition and observed follow-up domains in the v2 download. DSDY is the recorded safety-follow-up day, while LASTCTDY is the last-contact study day. Neither field is treated as a fixed-horizon outcome by itself.

| Arm                           | <i>n</i> | Completed safety follow-up | Mean DSDY | Mean LASTCTDY | LASTCTDY < 84 | LASTCTDY ≥ 84 |
|-------------------------------|----------|----------------------------|-----------|---------------|---------------|---------------|
| Chemotherapy                  | 260      | 181 (69.6%)                | 110.7     | 376.8         | 47            | 213           |
| Panitumumab plus chemotherapy | 260      | 173 (66.5%)                | 159.5     | 403.5         | 41            | 219           |

## 5.4 Posterior Models and Computation

ZIC-BCF-Smear models the hurdle and positive components with separate Bayesian causal forests. The hurdle model yields posterior draws of the probability of a positive reconstructed burden, and the continuous model operates on the positive outcome. A smearing re-transformation returns the total-outcome contrast to the original event-day scale (Duan, 1983). Each outcome was fit as four independent chains of 4,000 retained draws after 5,000 burn-in iterations, with seeds 1 through 4, and the reported posterior is the pooled set of 16,000 draws. For every posterior draw, the analysis saves the unit-level total-outcome CATE and the hurdle-margin CATE. A subgroup posterior is the average of its members’ unit-level effects within each draw. A contrast posterior is formed by subtracting the two subgroup averages within draw. This preserves posterior covariance, rather than comparing unrelated marginal intervals.

The Gamma-hurdle benchmark uses the same treatment coding, analytic cohort, and dummy-encoded baseline covariates for each outcome, drawn from the same shared definition file so that the two specifications cannot drift apart. It models the positive outcome with a Gamma component and the zero versus positive component with a hurdle model. Each fit used four chains of 1,000 retained draws after 1,000 burn-in iterations, with the same four seeds. The Gamma fit also returns posterior hurdle probabilities, allowing direct comparison of the probability-difference margin. Flexible Bayesian models for zero-inflated outcomes have been developed precisely because both distributional features can matter for causal estimation and uncertainty propagation (Oganisian et al., 2021).

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Convergence diagnostics are computed by the analysis scripts and follow the construction of Section 2.5. On the summed-duration endpoint the largest Gelman–Rubin statistic across all monitored quantities is 1.001 for ZIC-BCF-Smear and 1.000 for the Gamma hurdle, the smallest effective sample sizes are 2,670 and 16,000, and the largest absolute Geweke statistics are 2.03 and 1.42. On the distinct-calendar-day endpoint the corresponding values are 1.002, 2,777, and 2.21 for ZIC-BCF-Smear and 1.000, 16,000, and 1.42 for the Gamma hurdle. The assessed Geweke set is large, so a plain reading of its maximum needs a multiplicity context: eighteen functionals across four chains give 72 standard normal statistics per fit, whose 5% family-wise critical value is about 3.38, and the observed maxima of 2.03 and 2.21 sit well below it. The final fits use a 5,000-iteration burn-in because the originally reported 2,000-iteration configuration did not clear this bar: there the largest statistics were 3.67 on the summed-duration endpoint and 4.90 on the distinct-calendar-day endpoint, both attaching to  $\sigma_c$ , the log-scale residual standard deviation that sets the smearing factor, with family-wise probabilities of about 0.017 for the summed-duration endpoint and about  $7 \times 10^{-5}$  for the distinct-calendar-day endpoint under the correction. An investigation by direct replication, archived with the analysis scripts, established both the mechanism and the remedy. Conditional on the forest fit,  $\sigma_c$  receives a conjugate inverse-chi-square update, so its individual draws are nearly uncorrelated (lag-one autocorrelation about 0.04, effective sample size about 93% of nominal), which is why effective sample size and Gelman–Rubin diagnostics look clean throughout; but the joint level of  $\sigma_c$  and the fitted log-scale surface equilibrates through a two-way Gibbs coupling whose relaxation time varies by chain, so at the shorter burn-in some retained chains carried a small decaying level transient. Across twenty-four fresh seed combinations at that configuration the per-chain Geweke statistics of  $\sigma_c$  had standard deviation about 1.46 against the calibrated null value of 1, and every window pairing touching the early portion of a flagged chain flagged together. Extending burn-in to 5,000 iterations on the identical seeds removed every family-wise-threatening value while moving posterior means by at most 0.15%, orders of magnitude below reported precision, so the longer burn-in was adopted for all final models. The two remaining diagnostics cannot detect such transients by construction:  $\hat{R}$  pools split-half means across chains, against which one chain’s early tilt is negligible, and effective sample size measures variation around a fixed level, which a one-off decaying shift does not generate, leaving the early-versus-late Geweke contrast as the only of the three sensitive to residual burn-in. Because the quantity involved is the one that sets the smearing factor, Section 2.6’s locally smeared sensitivity analysis remains the relevant robustness check, and it bounds the consequence of exactly this kind of residual-scale behavior.

The homoskedasticity assumption behind the smearing factor is tested as described in Section 2.6. The studentized Breusch–Pagan test gives  $p = 0.122$  on summed duration and  $p = 0.157$  on distinct calendar days. The locally smeared sensitivity analysis moves the summed-duration ATE from 14.54 to 11.70 event-days, a 19.5% reduction that remains within the scalar-smeared posterior interval, and moves the distinct-day ATE from 9.75 to 9.39, a 3.7% reduction. The conclusions of this study are therefore not carried by the scalar smearing assumption.

Both models retain a credible positive effect when overlapping events are counted once. Relative to summed duration, the ZIC-BCF-Smear ATE decreases by 4.27 days

Table 12: Posterior average treatment effects. Positive values favor a higher burden or a higher probability of a positive reconstructed severe-AE burden under panitumumab assignment.

| Model and estimand                                                | Estimate     | 95% credible interval | $P(> 0)$ |
|-------------------------------------------------------------------|--------------|-----------------------|----------|
| ZIC-BCF-Smear, summed duration (event-days)                       | 14.27        | [4.69, 24.23]         | 0.999    |
| Gamma Hurdle, summed duration (event-days)                        | 15.95        | [6.15, 26.30]         | 0.999    |
| <i>Unadjusted arm difference, summed duration</i>                 | <i>15.68</i> | –                     | –        |
| ZIC-BCF-Smear, distinct calendar days                             | 10.00        | [3.57, 16.62]         | 0.999    |
| Gamma Hurdle, distinct calendar days                              | 12.20        | [5.56, 19.23]         | 1.000    |
| <i>Unadjusted arm difference, distinct calendar days</i>          | <i>12.06</i> | –                     | –        |
| ZIC-BCF-Smear, positive reconstructed burden, summed-duration fit | 0.0288       | [−0.0379, 0.1019]     | 0.786    |
| Gamma Hurdle, positive reconstructed burden, summed-duration fit  | 0.0171       | [−0.0554, 0.0893]     | 0.683    |
| ZIC-BCF-Smear, positive reconstructed burden, distinct-day fit    | 0.0273       | [−0.0392, 0.1023]     | 0.774    |
| Gamma Hurdle, positive reconstructed burden, distinct-day fit     | 0.0171       | [−0.0554, 0.0893]     | 0.683    |

and the Gamma-hurdle ATE decreases by 3.83 days. This expected attenuation reflects the change in endpoint rather than a reversal of the safety conclusion. The probability-margin estimates remain uncertain and almost unchanged because the zero indicator is unchanged. Thus, the supported conclusion concerns accumulated recorded burden, not a clearly established change in the fraction of patients who ever experience a severe AE. The Gamma-hurdle benchmark now uses the same four-chain, 4,000-draw-per-chain budget as ZIC-BCF-Smear on both endpoints, and every estimate above is within Monte Carlo error of the shorter benchmark run, so the specification comparison no longer mixes unequal posterior sample sizes.

Both models also sit close to the unadjusted arm differences, and under randomization that is the benchmark rather than a weakness. Because assignment was randomized, the raw contrast is itself an unbiased estimate of the average treatment effect in this cohort, so a fitted model whose point estimate departed sharply from it would owe the reader an explanation. The Gamma-hurdle distinct-day estimate of 12.20 days against an unadjusted 12.06 is therefore reassuring rather than uninformative. What the adjusted fits contribute is not bias removal, which randomization already supplies, but precision, the decomposition into participation and intensity, and the resolution of the effect by covariate profile. This point was previously stated only in passing and only in the supplement, and it is now stated in the main text as well.

An independent Gamma-hurdle fit of the distinct-day endpoint with seed 2026, outside the four seeds used by the main analysis, is not used for the updated primary summary. The four-chain result above has  $\hat{R} = 1.000$  and a stable positive posterior, which is sufficient for the reported specification comparison.

Figure 9 displays the two posterior distributions of the distinct-calendar-day ATE that underlie the third and fourth rows of Table 12. Three features are worth noting. First, both posteriors place essentially all of their mass to the right of zero, which is the graphical counterpart of  $P(\text{ATE} > 0) = 0.999$  for ZIC-BCF-Smear and 1.000 for the Gamma hurdle; the zero reference sits in the extreme left tail rather than outside the plotted support, so the evidence is strong without being degenerate. Second, the ZIC-BCF-Smear posterior is centred lower (10.00 days) and is mildly right-skewed, with a longer upper tail extending beyond 20 days. That shape is expected from the smearing

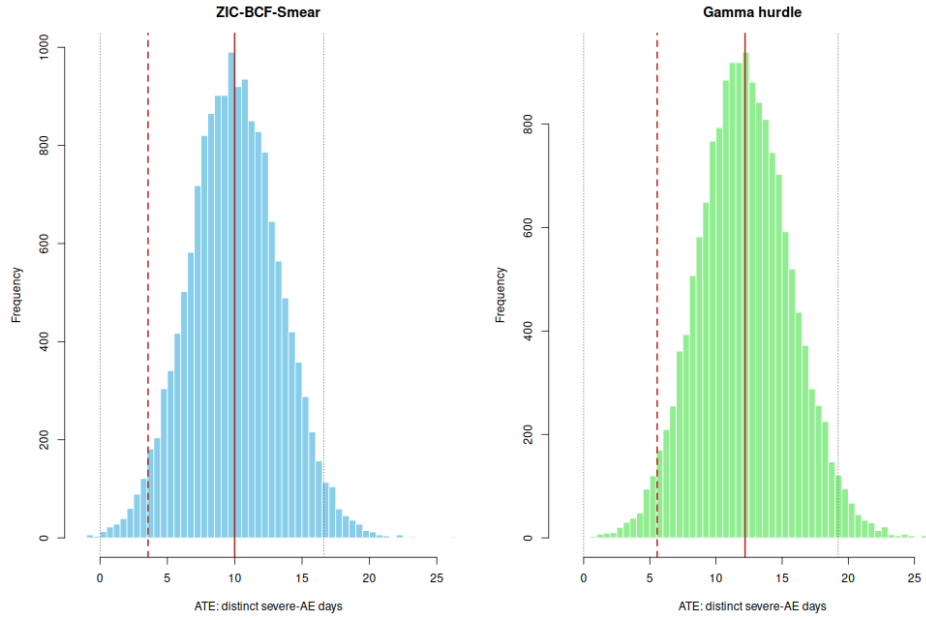


Figure 9: Posterior distributions of the average treatment effect on distinct severe-AE calendar days under ZIC-BCF-Smear (left) and the Gamma-hurdle benchmark (right). In each panel the solid vertical line marks the posterior mean, the dashed line the 2.5% posterior quantile, and the dotted lines the zero reference and the 97.5% posterior quantile.

re-transformation, which propagates uncertainty in the positive-part residual distribution onto the original day scale and therefore does not force a symmetric effect posterior. Third, the two posteriors overlap over almost their entire range, so the 2.15-day gap between the ZIC-BCF-Smear and Gamma-hurdle point estimates is small relative to the width of either distribution and should not be read as a substantive disagreement about the average effect.

## 5.5 Subgroup Results and Benchmark Sensitivity

Table 13 reports the ZIC-BCF-Smear CATEs. The subgroup effects are all positive in posterior mean, but the width of the intervals grows in small strata. No pairwise total-outcome contrast has a 95% credible interval excluding zero. The strongest ZIC-BCF-Smear signals, ECOG 0 versus ECOG 1 and HPV-negative versus HPV-not-recorded, remain compatible with no contrast.

Table 13:: ZIC-BCF-Smear subgroup CATEs for summed severe-AE-record duration.

| Covariate          | Level                 | <i>n</i> | CATE  | 95% credible interval | <i>P</i> (> 0) |
|--------------------|-----------------------|----------|-------|-----------------------|----------------|
| Overall            | All                   | 520      | 14.27 | [4.69, 24.23]         | 0.999          |
| Sex                | Male                  | 455      | 14.20 | [4.53, 24.37]         | 0.999          |
| Sex                | Female                | 65       | 14.76 | [2.28, 29.89]         | 0.990          |
| ECOG status        | ECOG 0                | 150      | 18.74 | [5.76, 35.51]         | 1.000          |
| ECOG status        | ECOG 1                | 370      | 12.46 | [3.27, 22.47]         | 0.996          |
| Diagnosis site     | Hypopharynx           | 75       | 13.37 | [3.56, 24.47]         | 0.997          |
| Diagnosis site     | Larynx                | 157      | 14.00 | [3.49, 26.50]         | 0.997          |
| Diagnosis site     | Oral cavity           | 142      | 14.57 | [3.22, 28.53]         | 0.995          |
| Diagnosis site     | Oropharynx            | 146      | 14.72 | [3.92, 27.77]         | 0.998          |
| HPV status         | Negative              | 270      | 16.85 | [5.88, 28.96]         | 1.000          |
| HPV status         | Positive              | 69       | 18.17 | [4.50, 36.73]         | 0.997          |
| HPV status         | Not recorded          | 181      | 8.93  | [−1.02, 19.27]        | 0.960          |
| Disease status     | Newly diagnosed       | 113      | 16.97 | [4.43, 33.48]         | 0.998          |
| Disease status     | Recurrent             | 407      | 13.52 | [4.14, 23.54]         | 0.999          |
| Prior radiotherapy | No prior radiotherapy | 104      | 16.87 | [4.10, 33.76]         | 0.997          |
| Prior radiotherapy | Prior radiotherapy    | 416      | 13.62 | [4.17, 23.65]         | 0.999          |
| Age group          | < 50                  | 72       | 15.83 | [4.87, 29.76]         | 0.999          |
| Age group          | 50–59                 | 240      | 13.92 | [4.45, 23.90]         | 0.999          |
| Age group          | 60–69                 | 172      | 13.92 | [4.35, 24.31]         | 0.998          |
| Age group          | 70+                   | 36       | 15.15 | [4.51, 27.95]         | 0.998          |

The Gamma-hurdle benchmark is deliberately included as a specification sensitivity analysis, not as a selector of a clinically preferred subgroup claim. Its total-duration ECOG 0 minus ECOG 1 contrast is 26.94 event-days (95% credible interval [4.74, 51.94]), whereas the corresponding ZIC-BCF-Smear contrast is 6.28 event-days (95% credible interval [−2.57, 22.20]). The cross-model disagreement remains substantive, and prevents a robust claim that baseline ECOG modifies the total event-day burden.

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The models also disagree on certain hurdle-margin contrasts. Gamma Hurdle estimates ECOG 0 minus ECOG 1 at 0.169 (95% credible interval [0.014, 0.323]) and HPV-negative minus HPV-not-recorded at 0.224 (95% credible interval [0.066, 0.387]). The corresponding ZIC-BCF-Smear contrasts are 0.041 (95% credible interval [−0.026, 0.176]) and 0.067 (95% credible interval [−0.020, 0.238]), and no ZIC-BCF-Smear hurdle-margin contrast excludes zero. Since the average hurdle effect is uncertain for both models, these isolated benchmark-dependent findings are hypothesis-generating only.

#### Two Contrasts That Need Naming

Two of the contrasts in these tables should be described for what they measure rather than for the covariate they are labeled with.

The HPV-not-recorded category is created by coalescing patients with no recorded baseline HPV result, and it contains 181 of the 520 patients. Any contrast against it is therefore a missingness contrast, not a comparison of two measured biological states, and the HPV-negative minus HPV-not-recorded contrast that the Gamma benchmark reports as credible on the participation margin is a statement about which patients had a recorded result rather than about HPV biology. It is retained in the tables because dropping the category would silently drop those 181 patients from the cohort, but it should not be read as evidence about human papillomavirus.

Prior radiotherapy is the clinically motivated candidate modifier, and it behaves like a candidate rather than an established one. On the summed-duration endpoint the effect is larger among the 104 patients without prior radiotherapy (16.87 event-days, 95% credible interval [4.10, 33.76]) than among the 416 with prior radiotherapy (13.62, [4.17, 23.65]), with a contrast of 3.25 event-days (95% credible interval [−7.61, 17.82]); on the distinct-day endpoint the corresponding estimates are 13.09 and 9.22 days, with contrast 3.87 days (95% credible interval [−4.10, 15.62]). These contrasts are not credible, and the smaller stratum is limited. The composite PRHNTRTC differs from PRRADIO for only four patients, all in the chemotherapy arm, so its subgroup contrast is a near-duplicate of the one reported here; it is omitted from the prespecified lists because it would add an unadjusted comparison that carries no additional information. The radiotx domain confirms PRRADIO exactly at the subject level for all 520 patients, none of them absent from the domain; the lesion-level flag is not a substitute because it is lesion-specific and, as documented above, contradicts PRRADIO for 42 of the 104 unirradiated patients.

Figure 10 presents the same estimates graphically and makes the shrinkage behaviour of the causal forest visible. Every subgroup interval lies to the right of zero except HPV-not-recorded, and every posterior mean falls within roughly five event-days of the overall ATE, so the ZIC-BCF-Smear fit describes a broadly homogeneous effect whose apparent subgroup variation is small relative to within-subgroup uncertainty. Interval width tracks subgroup size almost monotonically, with the widest intervals for HPV-positive patients ( $n = 69$ ), women ( $n = 65$ ), and patients aged 70 years or older ( $n = 36$ ), which is the expected consequence of borrowing strength across a tree ensemble rather than fitting separate stratum-specific models.

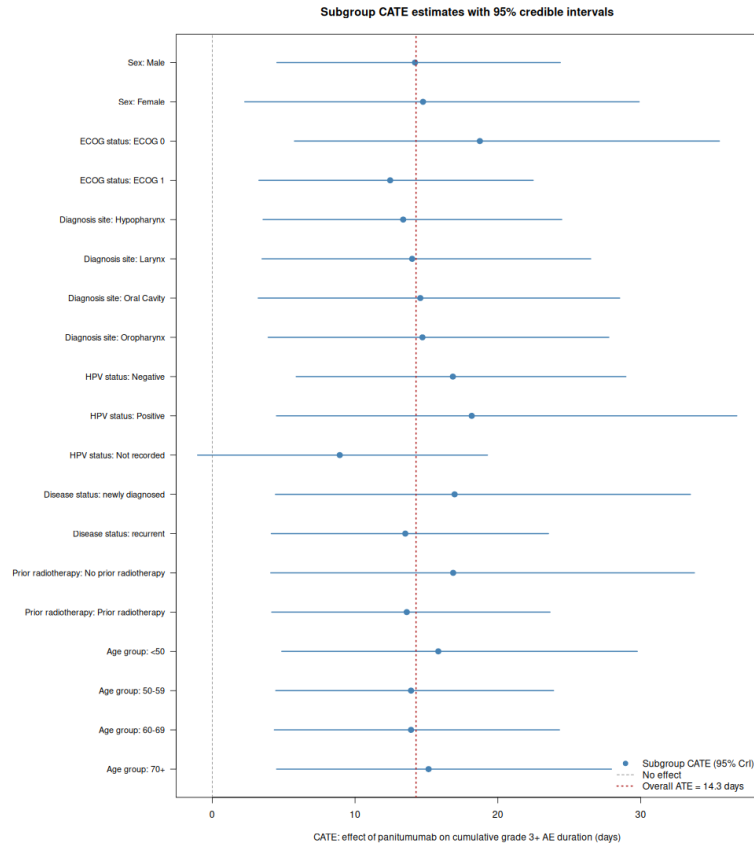


Figure 10: ZIC-BCF-Smear subgroup CATEs for summed severe-AE-record duration, with 95% credible intervals. The dashed vertical line at zero is the no-effect reference and the dotted vertical line is the overall ATE of 14.3 event-days.

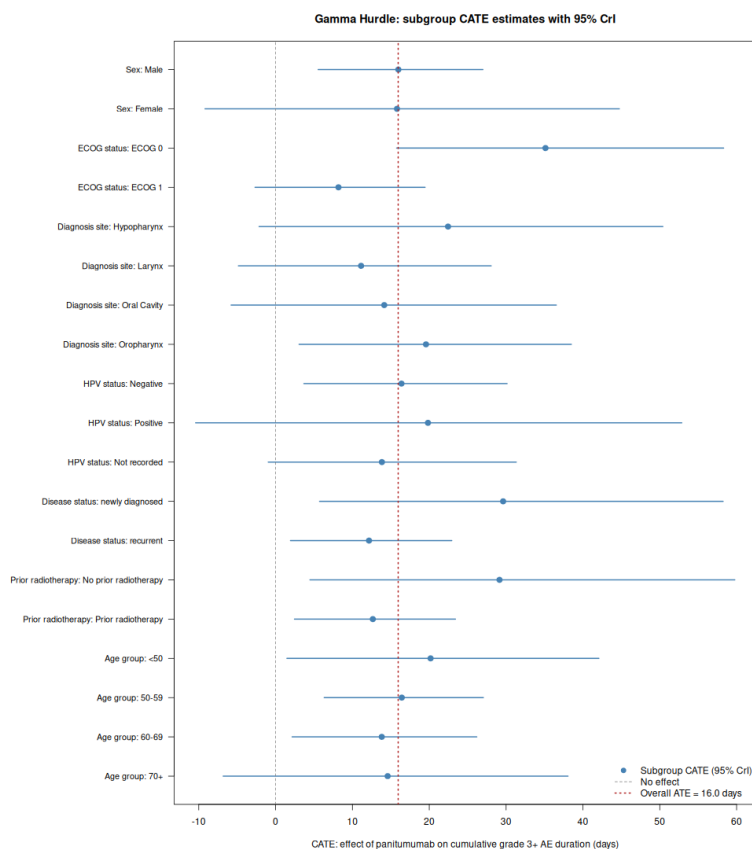


Figure 11: Gamma-hurdle subgroup CATEs for the same summed severe-AE-record duration outcome, on the same 95% credible-interval scale. The dotted vertical line is the Gamma-hurdle overall ATE of 16.1 event-days. Note that the horizontal axis spans roughly twice the range of Figure 10.

Figure 11 shows the corresponding Gamma-hurdle panel and, read against Figure 10, isolates where the two specifications actually part company. The Gamma-hurdle estimates are more dispersed: several intervals now cross zero (women, hypopharynx, larynx, oral cavity, HPV-positive, HPV-not-recorded, and patients aged 70 years or older), and the ECOG contrast is driven by an ECOG 0 posterior mean of 35.2 event-days against 8.3 event-days for ECOG 1, a separation with no counterpart in the ZIC-BCF-Smear panel. Because both models are fit to identical data, covariates, and treatment coding, the difference is attributable to the specification: the parametric Gamma positive component with linear dummy-encoded covariates extrapolates subgroup means without the regularization that the causal forest imposes. The comparison therefore supports the interpretation of the ECOG signal as specification sensitivity rather than as corroborated effect modification.

### Distinct-Calendar-Day Subgroup Sensitivity

Table 14 reports the ZIC-BCF-Smear CATEs for the overlap-adjusted endpoint. Posterior means are positive in every prespecified subgroup, but no between-group contrast has a 95% credible interval excluding zero. The largest contrast is ECOG 0 minus ECOG 1 at 6.69 days (95% credible interval  $[-0.83, 20.43]$ ). The corresponding Gamma-hurdle contrast is 23.34 days (95% credible interval  $[7.83, 40.98]$ ). Because the ZIC-BCF-Smear interval includes zero, this apparent ECOG heterogeneity remains model-dependent and should not be presented as a substantive subgroup finding.

Table 14:: ZIC-BCF-Smear subgroup CATEs for distinct severe-AE calendar days.

| Covariate          | Level                 | <i>n</i> | CATE  | 95% credible interval | $P(> 0)$ |
|--------------------|-----------------------|----------|-------|-----------------------|----------|
| Overall            | All                   | 520      | 10.00 | [3.57, 16.62]         | 0.999    |
| Sex                | Male                  | 455      | 10.12 | [3.55, 17.01]         | 0.999    |
| Sex                | Female                | 65       | 9.10  | [0.40, 18.63]         | 0.980    |
| ECOG status        | ECOG 0                | 150      | 14.76 | [4.86, 28.25]         | 1.000    |
| ECOG status        | ECOG 1                | 370      | 8.07  | [1.85, 14.58]         | 0.994    |
| Diagnosis site     | Hypopharynx           | 75       | 9.55  | [2.65, 17.22]         | 0.997    |
| Diagnosis site     | Larynx                | 157      | 10.88 | [3.15, 20.34]         | 0.998    |
| Diagnosis site     | Oral cavity           | 142      | 9.46  | [1.82, 18.30]         | 0.993    |
| Diagnosis site     | Oropharynx            | 146      | 9.80  | [2.37, 18.36]         | 0.997    |
| HPV status         | Negative              | 270      | 11.92 | [4.36, 20.05]         | 1.000    |
| HPV status         | Positive              | 69       | 12.22 | [3.17, 23.74]         | 0.997    |
| HPV status         | Not recorded          | 181      | 6.28  | $[-0.73, 13.50]$      | 0.961    |
| Disease status     | Newly diagnosed       | 113      | 13.02 | [3.51, 25.67]         | 0.999    |
| Disease status     | Recurrent             | 407      | 9.16  | [2.89, 15.61]         | 0.999    |
| Prior radiotherapy | No prior radiotherapy | 104      | 13.09 | [3.40, 26.02]         | 0.998    |
| Prior radiotherapy | Prior radiotherapy    | 416      | 9.22  | [2.97, 15.66]         | 0.999    |
| Age group          | < 50                  | 72       | 11.70 | [3.86, 21.60]         | 0.999    |
| Age group          | 50–59                 | 240      | 9.85  | [3.41, 16.54]         | 0.999    |
| Age group          | 60–69                 | 172      | 9.50  | [3.09, 16.26]         | 0.999    |

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| Covariate | Level | <i>n</i> | CATE | 95% credible interval | <i>P</i> ( $> 0$ ) |
|-----------|-------|----------|------|-----------------------|--------------------|
| Age group | 70+   | 36       | 9.93 | [2.81, 18.24]         | 0.997              |

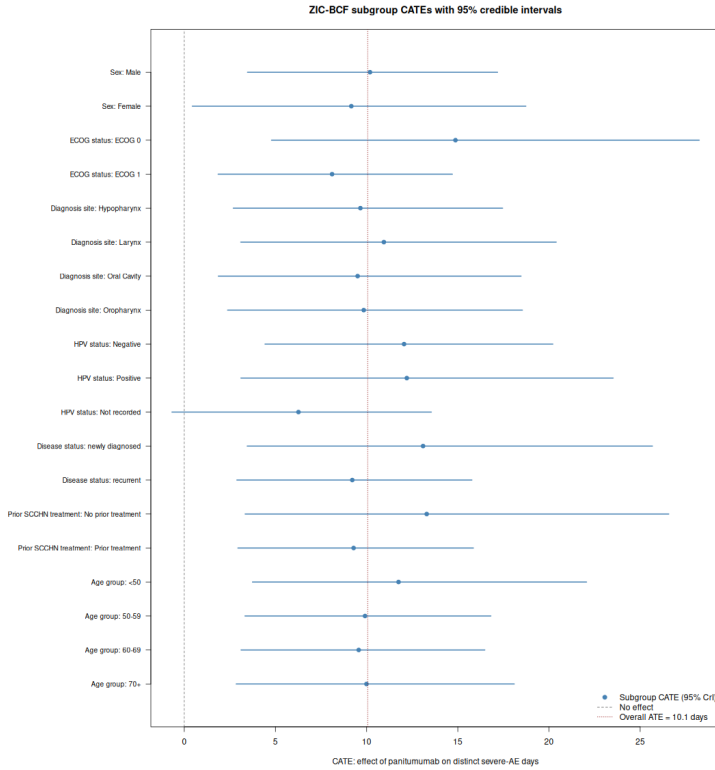


Figure 12: ZIC-BCF-Smear subgroup CATEs for distinct severe-AE calendar days, with 95% credible intervals. The dashed vertical line at zero is the no-effect reference and the dotted vertical line is the overall ATE of 10.1 days.

Figure 12 plots these estimates and should be compared directly with Figure 10. The two panels have the same ordering of subgroups and nearly the same shape, but the distinct-day axis is compressed by roughly a third, which is the graphical signature of counting each calendar day once. No subgroup changes sign, and the only interval that still covers zero is HPV-not-recorded, exactly as in the summed-duration fit. The ECOG 0 estimate remains the largest and the widest, and its interval visibly overlaps the ECOG 1 interval, which is why the ECOG contrast in this endpoint also fails to exclude zero. Taken together, the two figures show that the overlap adjustment rescales the ZIC-BCF-Smear effect surface without reorganizing it, so the conclusions about heterogeneity are not artefacts of double-counting concurrent events.

## 5.6 Position in the Clinical and Statistical Literature

The completed literature review supports four carefully separated statements. First, the direction of the average burden contrasts is clinically coherent. The primary SPECTRUM report found that the panitumumab regimen had more severe toxicity in several conventional safety categories (Vermorken et al., 2013). Second, neither numerical ATE is directly comparable with a published SPECTRUM effect, because standard safety tables summarize event incidence, grade, and category rather than summed record durations or overlap-adjusted calendar days. Third, previous SPECTRUM subgroup discussions focused principally on efficacy and HPV or p16 status; the available HPV-specific safety analysis is a conference abstract and does not estimate baseline-modified causal effects on either burden outcome (Stöhlmacher-Williams et al., 2012). Fourth, the review identified no prior study conducting the present secondary causal analysis of the SPECTRUM data.

The novel contribution is therefore specific and clinically interpretable. It combines 1. a patient-centred endpoint defined as distinct calendar days with at least one grade 3+ AE rather than AE counts or summed record durations, 2. an ATE of randomized panitumumab assignment estimated on that day scale, 3. a direct sensitivity analysis showing that the credibly positive overall conclusion persists after concurrent events are not double-counted, and 4. flexible, uncertainty-aware subgroup-effect estimation for a semicontinuous outcome. To our knowledge, this is the first secondary causal analysis of the SPECTRUM data using an overlap-adjusted severe-AE-day endpoint. This claim does not recast the known clinical direction as a new discovery and does not claim priority over every cumulative-toxicity analysis in oncology. The method is motivated by a broad statistical literature on semicontinuous outcomes and flexible causal modeling (Liu et al., 2019; Oganisian et al., 2021).

Against this background, the primary question is how far the ZIC-BCF-Smear results themselves converge with what the clinical literature already supports. The convergence is strongest, and is essentially complete, for the direction and stability of the average burden effect. ZIC-BCF-Smear estimates 14.27 additional event-days (95% credible interval [4.69, 24.23],  $P(> 0) = 0.999$ ) on the summed-duration scale and 10.00 additional days (95% credible interval [3.57, 16.62],  $P(> 0) = 0.999$ ) on the overlap-adjusted calendar-day scale. Both sign and near-certainty match the qualitative SPECTRUM safety finding that the panitumumab-containing regimen produced more severe toxicity (Vermorken et al., 2013), and the conclusion survives the one outcome-definition change that could plausibly have manufactured it, namely the double-counting of concurrent severe events. What the literature does not provide is a numerical benchmark: no published SPECTRUM analysis reports an effect on either duration scale, so external agreement can be assessed for direction and for robustness, but not for magnitude. The 4.27-day attenuation between the two endpoints is best read as a statement about how much of the summed-duration burden reflects simultaneity, and it is internal evidence rather than a replication of any published quantity.

Convergence is weaker, though not contradictory, at the level of mechanism. ZIC-BCF-Smear finds no credible population-average effect on the probability of a positive

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reconstructed severe-AE burden, estimating a 2.88 percentage-point increase (95% credible interval  $-3.79$  to  $10.19$ ) from the summed-duration fit and 2.73 percentage points (95% credible interval  $-3.92$  to  $10.23$ ) from the distinct-day fit. That margin is not literally the probability of experiencing any severe adverse event, since 13 patients with severe events enter the data with a zero burden because their observation intervals were incomplete. The published safety evidence emphasized particular adverse-event categories and grades rather than an aggregate any-severe-AE indicator, so there is no literature quantity that this margin either confirms or refutes. The substantive implication is internal to the model decomposition: the credible effect is carried by the positive component, that is, by how much severe-AE time accumulates among patients whose reconstructed burden is positive, rather than by the hurdle component that governs whether that burden is positive at all. Since 76.3% of the cohort has a positive burden in both arms, the hurdle margin has limited room to move, and the supported claim concerns positive intensity rather than occurrence. That intensity can reflect event duration, event multiplicity, overlap, treatment exposure, and incomplete interval reconstruction, so it should not be labeled persistence specifically.

Convergence is weakest for effect modification, and here the ZIC-BCF-Smear results are best described as literature-consistent but uninformative. Posterior means are positive in every prespecified subgroup on both endpoints, no pairwise contrast excludes zero, and the strongest signals, ECOG 0 versus ECOG 1 and HPV-negative versus HPV-not-recorded, remain compatible with no contrast. The SPECTRUM literature does not establish either ECOG or HPV status as a modifier of the causal burden of panitumumab-associated severe toxicity, and the available HPV-specific analysis is a conference abstract that does not analyze either duration endpoint ([Stöhlmacher-Williams et al., 2012](#)). A regularized model returning no credible heterogeneity is therefore concordant with the absence of established modifiers.

The Gamma-hurdle benchmark converges with the clinical literature on the same principal average-effect result. Its estimated effect of 15.95 event-days (95% credible interval  $[6.15, 26.30]$ ) for summed record durations, and 12.20 days (95% credible interval  $[5.56, 19.23]$ ) for distinct calendar days, agree in sign and broad magnitude with the corresponding ZIC-BCF-Smear estimates, and its any-severe-AE margin of 1.71 percentage points (95% credible interval  $-5.54$  to  $8.93$ ) is likewise uncertain. Agreement between a flexible nonparametric and a parametric specification on both endpoints strengthens the average-effect conclusion, since it is not an artefact of either functional form. The Gamma-hurdle point estimates are close to the unadjusted arm differences, and under randomization that is the internal benchmark a model should meet rather than a shortcoming: the raw contrast is itself unbiased for the cohort average effect. It is not external validation, since the published trial report did not analyze either duration endpoint, but neither is it a weakness of the fit.

The Gamma-hurdle subgroup findings are less literature-convergent, and this is where the two specifications diverge. For summed duration, Gamma Hurdle estimates an ECOG 0 minus ECOG 1 contrast of 26.94 event-days (95% credible interval  $[4.74, 51.94]$ ), while the corresponding ZIC-BCF-Smear contrast of 6.28 event-days has a credible interval that includes zero. For distinct calendar days, the contrast is 23.34 days

(95% credible interval [7.83, 40.98]) under Gamma hurdle and 6.69 days (95% credible interval [−0.83, 20.43]) under ZIC-BCF-Smear. Gamma Hurdle also gives positive hurdle-margin contrasts for ECOG 0 versus ECOG 1 and HPV-negative versus HPV-not-recorded, whereas the corresponding ZIC-BCF-Smear contrasts remain uncertain. Given that the clinical literature establishes neither modifier, these Gamma-specific signals are best interpreted as model-sensitive hypotheses rather than as externally corroborated clinical effect modification.

## 5.7 Limitations

The first limitation is outcome reconstruction. The sum in Equation (5.1) can double count concurrent events, and 7.5% of severe AE records lack its duration field. Equation (5.2) directly addresses overlap and confirms the positive overall conclusion, but 124 qualifying records have an unresolved end boundary, affecting 88 patients, and 13 patients with no complete qualifying interval are assigned zero outright with a 10-to-3 imbalance against panitumumab, as detailed in Section 5.2. The v2 disposition domains provide DSDY and LASTCTDY for every patient, but these do not identify the missing event end dates; the companion endpoint should therefore be interpreted as observed distinct calendar days represented by complete intervals, rather than total toxicity time through a common follow-up horizon. Neither endpoint silently replaces the other because they answer different clinical questions.

The second limitation is differential treatment exposure, and it is the most consequential one. Mean time on treatment is 56% longer under panitumumab, structurally so because the drug continues as maintenance after chemotherapy stops, and truncating accrual at a common study day collapses the unadjusted arm difference from 12.06 days to about 3 days. Section 5.3 develops this in full. The estimand is coherent, and under randomization it is identified, but it is the effect of assignment under the trial's own differential duration and not a per-exposure or fixed-horizon quantity. Because treatment duration is post-randomization it can be neither ignored in interpretation nor conditioned on in estimation, and the per-exposure analyses are reported as descriptive for that reason. A larger accumulated burden may therefore arise through a mixture of more toxicity, longer event duration, more co-occurring events, incomplete intervals, and more time on treatment, and this analysis does not separate those channels.

The third limitation is incomplete follow-up ascertainment rather than a total absence of follow-up fields. The v2 disposition file supplies DSDY and LASTCTDY for all 520 patients, but only 354 completed the safety follow-up, and the meaning of last contact is not identical to a completed safety assessment. The analysis therefore does not standardize follow-up, model competing death or progression, or estimate a censoring-adjusted fixed-horizon endpoint. A prespecified fixed-horizon restricted-mean or multistate analysis is now feasible in principle, but it would require an explicit censoring rule, treatment policy, and competing-event definition. The present manuscript reports the last-contact-capped endpoint only as a descriptive diagnostic.

The fourth limitation concerns the analytic cohort. Randomization licenses the causal reading, and randomization attaches to the parent trial's 657 patients rather

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than automatically to the 520-person extract analyzed here. Section 5.1 sets out what the supplied files do and do not establish about that reduction; the 520-to-514 reconciliation is complete, and the 657-to-520 step is not documented in any supplied file.

The fifth limitation concerns heterogeneity. Several subgroups are small, especially women, HPV-positive patients, patients aged 70 years or older, and the 100 patients with no prior head-and-neck treatment. The subgroup comparisons are exploratory and are not multiplicity-adjusted. Two of them measure something other than the covariate they are named for or rest on a stratum too small to resolve, as Section 5.5 sets out. And the Gamma-hurdle versus ZIC-BCF-Smear disagreement for the ECOG contrast persists under the distinct-day endpoint. This is useful sensitivity evidence, but it reinforces that no baseline effect-modification claim should be based on one model alone.

The sixth limitation concerns the resolution of prior-treatment information. PRHN-TRTC remains a composite of prior radiotherapy and other prior treatment, but the v2 download supplies the patient-level PRRADIO indicator and the detailed radiotx domain. PRRADIO agrees exactly with the subject-level radiotherapy records and is now used in the adjustment set and subgroup lists. The lesion-level LSPIRRNY flag contradicts PRRADIO for 42 of the 104 patients recorded as not irradiated (Section 5.1), so it cannot be used to recover a more granular patient-level radiation history until that discrepancy is resolved. Radiation-site and lesion-specific effect modification therefore remains an extension rather than an established result.

### 5.8 Reproducibility

All paths below are relative to the `applied_study_oncology/` directory of the analysis repository accompanying the `zicbcf` package (Souto, 2026). The upstream reconstruction is documented in `eadNe_265_severe_AE_screening.Rmd`. The de-identified analytic file, which now carries prior head-and-neck treatment, prior radiotherapy, treatment duration, and disposition fields, is rebuilt by `prepare_analysis_data.R`; the v2 domain reconciliation is run by `New_Data_v2/run_domain_diagnostics.R`. The adjustment set and subgroup definitions shared by every fit are defined once in `oncology_common.R`; the randomized propensity is supplied as 0.5. The ZIC-BCF-Smear fit is run by `run_zicbcf_oncology.R`, which also writes the convergence and smearing diagnostics; its hurdle-margin subgroup analysis is run by `run_zicbcf_oncology_hurdle_subgroups.R`. The Gamma-hurdle analysis is run by `run_gamma_hurdle_oncology.R` and `run_gamma_hurdle_oncology_subgroups.R`. The distinct-day endpoint is constructed by `New_Data/build_nonoverlap_calendar_day_endpoint.R` and the matched model fits by `New_Data/run_nonoverlap_calendar_day_analysis.R`, with forest plots drawn without refitting by `New_Data/plot_nonoverlap_zic_cate_figures.R`. The exposure, cohort-reconciliation and missingness diagnostics of Sections 5.1, 5.2 and 5.3 are produced by `New_Data/run_exposure_and_missingness_diagnostics.R` and `New_Data_v2/run_domain_diagnostics.R`; the descriptive person-time models are produced by `New_Data/run_exposure_rate_descriptive_analysis.R`.

Machine-readable endpoint data, posterior draws, model summaries, subgroup results, and figures are stored in `New_Data/analysis` and `New_Data_v2/analysis`.

## 5.9 Data Sharing

The individual-participant data analyzed in this section cannot be redistributed, and the reason is worth stating precisely because the usual intuition does not apply. Section III.B(ii) of the Project Data Sphere Platform Agreement prohibits disclosure of the data to any third party who is not a registered Authorized User, and Article VIII requires that they be held in strict confidence ([Project Data Sphere, 2020](#)). This is a contractual confidentiality obligation rather than a patient-privacy one, so the instinct that further de-identification makes a file safe to post does not automatically hold here.

We state the practical position rather than a legal one. The agreement does not expressly define the point at which a derived quantity ceases to be “the Data”, and Article VII does permit disclosure of what is reasonably necessary to publish or report research results, which is the basis on which the aggregate estimates, tables and figures in this section appear at all. Our operating rule is therefore that per-patient derivatives, including patient-level endpoint files and unit-level posterior draws, are not posted without written confirmation from Project Data Sphere or Amgen. The repository accordingly excludes every patient-level file and every per-patient derivative for this study, while committing all analysis scripts, so that a registered Authorized User can reproduce the study end to end from the supplied package.

Two consequences follow for the authorship and reporting of this work. Access is granted per individual, so every co-author who handled the patient-level data must be separately registered as an Authorized User. And Article VII makes an acknowledgement mandatory in prescribed wording; that sentence appears verbatim in the acknowledgements of the main manuscript and is not paraphrased. Two matters cannot be settled from outside the download record and should be confirmed against it: whether the accepted agreement is the same 2020 version cited here, and whether the Amgen 20050251 package carried dataset-specific Supplemental Terms, which Section III.A(ii) of the agreement permits.

## 6 Applied Study: Dental Insurance and Dental Expenditures in the Medical Expenditure Panel Survey

This section is the extended, self-contained version of the health-economics applied study referred to in the main manuscript. It develops in full the application to the 2023 Medical Expenditure Panel Survey: the data source and cleaning, the descriptive structure of the semicontinuous dental-expenditure outcome, the estimation pipeline and MCMC configuration, the draw-by-draw construction of subgroup effects, the complete set of subgroup CATE estimates and pairwise contrasts, and a detailed reconciliation of every finding against the applied dental health-economics literature, including the policy implications of the estimated heterogeneity. Because that literature is mature and quantitatively specific, this study functions as an external check on ZIC-BCF-Smear: it asks whether the estimator recovers gradients that decades of applied dental health-economics research have already established.

### 6.1 Data Source and Outcome Construction

#### The Medical Expenditure Panel Survey

The Medical Expenditure Panel Survey (MEPS) is a nationally representative survey of the U.S. civilian non-institutionalized population, administered by the Agency for Healthcare Research and Quality, and is a standard data source for studying healthcare utilization and expenditure. The 2023 Full-Year Consolidated File (HC-251) is used here ([Agency for Healthcare Research and Quality, 2023](#)). Dental expenditures recorded in MEPS are a canonical semicontinuous outcome and have been used repeatedly to study dental-coverage effects ([Manski et al., 2002](#); [Christian and Chattopadhyay, 2014](#); [Luo et al., 2021](#); [Naib et al., 2026](#)).

#### Variables

The analysis uses the following variables. The outcome  $Y$  is DVTEXP23, total annual dental expenditure in U.S. dollars, which is semicontinuous: a large point mass at exactly zero (non-utilizers) plus a heavily right-skewed continuous distribution among users. The treatment  $Z$  is DNTINS23\_M23, a dental-insurance coverage indicator coded 1 for covered and 0 otherwise. The covariates  $X$  are age, sex, race and ethnicity, family income, poverty category, census region, and marital status, which are standard determinants of dental utilization and expenditure in the MEPS literature ([Manski and Magder, 1998](#); [Christian and Chattopadhyay, 2014](#); [Zardak et al., 2023](#)). The design variables are the person weight PERWT23F, the variance stratum VARSTR, and the variance primary sampling unit VARPSU. After excluding invalid, negative, and inapplicable survey responses (for example, out-of-scope codes on the treatment and outcome fields), the causal analysis retains  $N = 18,763$  records.

Race and ethnicity are taken from RACETHX, the standard MEPS analytic variable, whose five levels are Hispanic, non-Hispanic White, non-Hispanic Black, non-Hispanic

Asian, and non-Hispanic other or multiple race. The first version of this study used RACEV2X, which records race only and carries no ethnicity dimension. In this sample that choice placed 3,758 Hispanic respondents inside the “White” category and 144 inside “Black”, so a racial gradient estimated on it was not measuring what its labels claimed. Since the racial gradient is the primary external check on the estimator, the variable has been changed and every affected number in this section has been recomputed.

### Encoding of Nominal Covariates

The design matrix is built with `model.matrix` over factors, so that each nominal covariate is expanded into indicator columns. This corrects a defect in the first version of the analysis, where the design matrix was formed by a direct numeric coercion of the covariate columns. Under that coercion race, poverty category, census region, and marital status entered the forests as single continuous columns whose values were arbitrary category codes. The consequence is not cosmetic: a split at `REGION23 ≤ 2` separates Northeast and Midwest from South and West purely because of the order in which the codes were assigned, and a split on race orders the categories along the code axis. The forests never saw race as a category, even though the reported race labels were constructed correctly for the output tables. The oncology scripts already used the correct convention, so the two applied studies in the same package were using incompatible ones; both now draw their design matrix from a single shared definition file.

### Descriptive Structure of the Outcome

The dental-expenditure outcome is exactly the regime that motivates a two-part, distribution-free treatment. In the analytic sample, 54.9% of respondents report zero dental spending, and 54.5% on a weighted basis. Among the positive (active) subpopulation, the distribution is severely right-skewed: the median positive expenditure is \$361, the mean is \$1,038, the 90th and 95th percentiles are \$2,505 and \$4,511, and the maximum is \$40,350. Figure 13 shows the zero-versus-positive decomposition, Figure 14 the approximately log-normal shape of the positive part on the log scale, and Figure 15 the raw association between coverage and the probability of any dental spending.

## 6.2 Estimation Pipeline

### Propensity Models

Two propensity models are fit. A full-sample logistic regression estimates  $\hat{\pi}(X) = P(Z = 1 \mid X)$ , which is supplied to the hurdle-stage prognostic forest. A second logistic regression is fit strictly on the active subpopulation  $\{i : Y_i > 0\}$  to estimate the subpopulation propensity  $\hat{\pi}^+(X) = P(Z = 1 \mid X, Y > 0)$ , which is supplied to the continuous-stage prognostic forest to implement the Subpopulation Propensity Adjustment (SPA). SPA is required because conditioning on the post-treatment event  $\{Y > 0\}$  opens a collider path; adjusting for  $\hat{\pi}^+$  within the active set restores identification of the intensity treatment effect under Assumption 6.1.

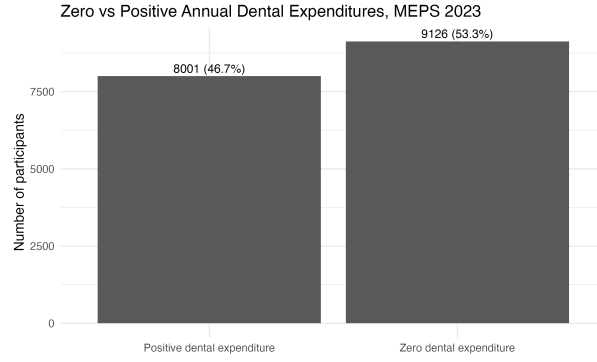


Figure 13: Decomposition of the dental-expenditure outcome into the zero point mass ( $\approx 53\%$  of records) and the positive continuous part.

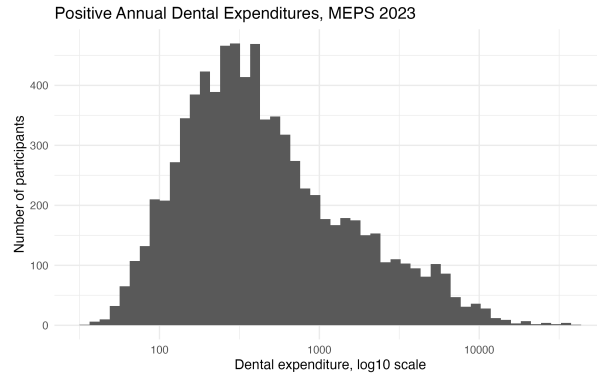


Figure 14: Histogram of positive dental expenditures on the log scale, showing the strong right skew on the raw scale and the approximate log-normality that motivates the log-link continuous stage combined with a nonparametric (smearing) re-transformation.

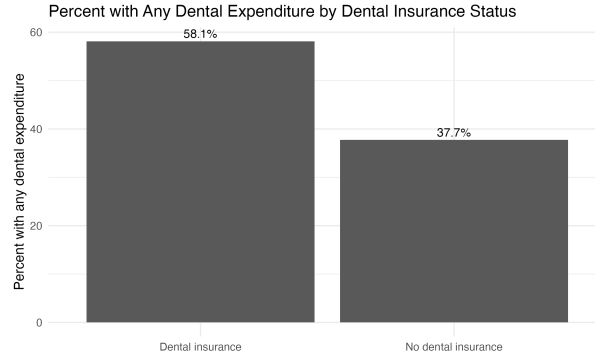


Figure 15: Unadjusted probability of any dental expenditure by dental-insurance status. Coverage is strongly associated with clearing the zero hurdle; the causal analysis adjusts this association for confounding and separates it from the intensity effect.

**Assumption 6.1** (Subpopulation strong ignorability). Conditional on covariates  $X$  and active status  $\{Y > 0\}$ , treatment  $Z$  is independent of the potential positive-intensity outcomes,  $(Y^+(0), Y^+(1)) \perp Z \mid X, Y > 0$ . As established in the main manuscript, this holds under full-sample strong ignorability together with conditional two-part independence of the participation and intensity processes.

### ZIC-BCF-Smear Configuration

The `zicbcf_smear` routine of the `zicbcf` package (Souto, 2026) is used. The hurdle stage is a Probit BCF fit on all  $N$  records (participation), and the continuous stage is a Gaussian BCF fit on  $\log(Y^+)$  over the active subset (intensity), each following the BCF construction of Hahn et al. (2020) with BART components (Chipman et al., 2010; Hill, 2011). Log-scale predictions are mapped to the dollar scale draw-by-draw with Duan’s nonparametric smearing factor (Duan, 1983), which avoids the parametric multiplier  $\exp(\sigma^2/2)$  and its misspecification bias (Manning and Mullahy, 2001). Four independent chains are run with seeds 1 through 4, each for 2,000 retained draws after 5,000 burn-in iterations, giving 8,000 pooled draws. Chains are run and summarized one at a time and then discarded, so that peak memory stays near the size of a single fit on an 18,763-record sample; what is retained is the per-draw average of unit-level effects within each reported domain, which is sufficient for every subgroup effect and every contrast reported below, together with the unit-level posterior means used by the marginal figures.

### Survey Design and the Two Reported Estimands

MEPS is a stratified, clustered, unequal-probability sample of the U.S. civilian non-institutionalized population (Agency for Healthcare Research and Quality, 2025). The

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analytic sample spans 105 variance strata and 262 primary sampling units, every stratum containing at least two, and the person weights sum to 331,946,662 persons. An unweighted average over the 18,763 records is therefore a sample quantity over an over-sampled design, and an interval built from the posterior alone ignores both stratification and clustering. The first version of this study reported only that quantity, and cut its income tertiles at unweighted quantiles.

Both estimands are now reported side by side for every effect, because they answer different questions and the difference between them is itself informative.

The *analytic-sample average* is the unweighted average of unit-level effects,  $|G|^{-1} \sum_{i \in G} \tau^{(s)}(X_i)$ , summarized by its posterior. It describes the sample.

The *survey-weighted population target* is the weighted average  $(\sum_{i \in G} w_i)^{-1} \sum_{i \in G} w_i \tau^{(s)}(X_i)$  with  $w_i = \text{PERWT23F}$ , taken within each posterior draw. Its interval adds a design-based variance component to the posterior variance. The design component is obtained by Taylor linearization (Binder, 1983; Wolter, 2007): for a domain mean the linearized value of unit  $i$  is  $u_i = w_i d_i (\tau_i - \hat{\theta}_G) / \sum_j w_j d_j$  with  $d_i$  the domain indicator, these are summed within primary sampling unit, and the variance is the usual between-PSU within-stratum sum of squares  $\sum_h \frac{n_h}{n_h - 1} \sum_{a \in h} (u_{ha} - \bar{u}_h)^2$ . Units outside a domain contribute zero but remain attached to their sampling unit, so domain estimation is handled correctly. For a between-domain contrast the linearized values of the two domains are differenced before aggregation, which retains the covariance induced by their sharing primary sampling units. The reported interval is the posterior mean plus or minus 1.96 times  $\sqrt{V_{\text{posterior}} + V_{\text{design}}}$ .

Two properties of this construction should be stated plainly. It treats sampling variability and posterior uncertainty as approximately independent, which is the standard decomposition when a model-based estimate is carried onto a complex-survey design but is an approximation rather than an exact result. And it does not let the design influence the fitted response surface: the sampler has no weight interface, so weighting acts only on the linear functionals of unit-level effects that are reported. Income tertile cutpoints are now weighted quantiles, which moves them substantially, from \$45,400 and \$110,600 unweighted to \$56,133 and \$125,320 weighted, and correspondingly changes which respondents fall in which tertile.

It is not knowable in advance whether respecting the design widens or narrows an interval, since clustering ordinarily widens and stratification ordinarily narrows. In this application the design component turns out to be small relative to the posterior: for the overall dollar-scale effect the design standard error is \$1.34 against a posterior standard deviation of \$15.73. That is a finding rather than an assumption, and it did not have to come out that way.

### Draw-by-Draw Subgroup and Contrast Estimation

All subgroup summaries are computed from the posterior draw matrix, so that uncertainty propagates correctly.

1. **Overall ATE.** For each draw  $s$ , average the unit-level effects over all  $N$  units,  $\text{ATE}^{(s)} = N^{-1} \sum_i \tau_{\text{resp}}^{(s)}(X_i)$ ; the posterior mean and 2.5/97.5% quantiles give the point estimate and 95% credible interval.
2. **Subgroup CATE.** For a subgroup  $G$ , average within each draw over its members only,  $\text{CATE}_G^{(s)} = |G|^{-1} \sum_{i \in G} \tau_{\text{resp}}^{(s)}(X_i)$ , giving a full posterior for the subgroup effect.
3. **Contrasts.** For two subgroups  $G_1, G_2$ , form the per-draw difference  $\Delta^{(s)} = \text{CATE}_{G_1}^{(s)} - \text{CATE}_{G_2}^{(s)}$ ; the posterior of  $\Delta$  yields a credible interval and the posterior probability  $P(\Delta > 0)$ .

This is the same logic used for the overall ATE, restricted to the units defining each subgroup, and it is what distinguishes a proper subgroup CATE analysis from inspecting the marginal spread of unit-level point estimates.

## 6.3 Results

### Average Treatment Effect

The estimated ATE of dental insurance on annual dental expenditure is \$188.02 as an analytic-sample average, with a 95% credible interval of [\$156.95, \$220.67], and \$184.67 as a survey-weighted population target, with a design-aware 95% interval of [\$154.63, \$214.70]. The posterior probability of a positive effect is 1.00 under both. Figure 16 shows the ATE posterior and Figure 17 the marginal distribution of the 18,763 unit-level posterior-mean CATEs.

Convergence diagnostics follow Section 2.5. Across the seven monitored functionals the largest  $\hat{R}$  is 1.007, the smallest effective sample size is 1,313 out of 8,000 pooled draws, and the largest absolute Geweke statistic is 2.80. The multiplicity context of Section 2.5 applies here as well: seven functionals across four chains give 28 standard normal statistics, whose 5% family-wise critical value is about 3.12, and the observed maximum of 2.80 sits below it, so no correction-flagged value remains. The maximum moved rather than merely shrank: under the longer burn-in  $\sigma_c$  falls from 3.672 to 0.556 while the hurdle effect rises from 0.899 to 2.804, so the surviving maximum is a different functional rather than a uniformly quieter posterior. The MEPS fits share the sampler core, and therefore the  $\sigma_c$  equilibration behavior, with the SPECTRUM fits; the final models accordingly use the same 5,000-iteration burn-in motivated there, which the investigation reported in that section showed removes the early-window transient that produced the originally flagged value. Because  $\sigma_c$  sets the smearing factor that rescales every dollar-scale summary, the locally smeared sensitivity analysis below remains the check that bounds what residual-scale behavior can do to the reported estimate. The Monte Carlo standard error of the dollar-scale ATE is \$0.40, two orders of magnitude below its posterior standard deviation, so the reported summaries are not limited by sampler error.

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The homoskedasticity assumption behind the smearing factor is tested as described in Section 2.6, and here it fails decisively. The studentized Breusch–Pagan statistic is 160.3 on 20 degrees of freedom,  $p = 6.4 \times 10^{-24}$ , and the draw-wise version rejects at the 1% level in all 200 draws examined. The residual standard deviation rises monotonically across the five strata of the fitted log-scale mean, from 1.005 in the lowest to 1.192 in the highest, which is the expected pattern for expenditure data. What matters for the reported estimate, however, is the locally smeared sensitivity analysis, and it moves the pooled-average effect only from \$188.02 to \$193.76, a change of 3.0% against a posterior standard deviation of \$16.2. The conclusion is therefore that the assumption is violated and that the violation does not materially drive the estimate, which are two separate findings and both worth reporting.

**Locally smeared subgroup sensitivity.** Because the smearing factor rises by roughly 30% between the extreme strata of the fitted mean, the quantity most exposed to that heterogeneity is not the cohort average, over which stratum-specific rescalings partly cancel, but the subgroup ordering on which any disparity statement rests. The locally smeared sensitivity analysis was therefore repeated for every reported subgroup: within each posterior draw, the single scalar factor is replaced by the smearing factor of the stratum each unit’s fitted log-scale mean falls into, the unit-level response-scale CATEs are recombined on that basis, and each subgroup posterior is recomputed draw-by-draw exactly as in Table 16. Both the locally smeared estimates and their scalar-smeared counterparts below come from the same pooled four-chain run, so the comparison isolates the re-transformation rather than Monte Carlo noise. Table 15 reports the result, ordered by sensitivity.

Table 15:: Locally smeared subgroup sensitivity for the dollar-scale analytic-sample average: scalar Duan versus locally smeared subgroup CATEs, ordered by relative change. Both columns derive from the same pooled posterior draws; every locally smeared point estimate lies inside its scalar-smear 95% credible interval.

| Domain                                                 | $n$   | Scalar CATE (\$) | Locally smeared | Local 95% Cr     |
|--------------------------------------------------------|-------|------------------|-----------------|------------------|
| Age group : 35-49                                      | 3468  | 175.27           | 183.95          | [150.47, 218.70] |
| Age group : 18-34                                      | 3352  | 169.06           | 176.81          | [142.45, 214.46] |
| Family income (weighted tertile) : High-income tertile | 5330  | 188.25           | 196.54          | [149.30, 242.36] |
| Marital status (detailed) : Never married              | 4579  | 191.74           | 199.91          | [160.15, 246.77] |
| Region : Midwest                                       | 3837  | 195.04           | 202.46          | [153.53, 259.09] |
| Poverty cat. : High-income                             | 7351  | 206.16           | 213.89          | [164.10, 260.06] |
| Race and ethnicity : Asian, non-Hispanic               | 1143  | 208.13           | 215.66          | [156.84, 287.64] |
| Marital status : Married                               | 7497  | 199.66           | 206.47          | [163.18, 248.00] |
| Marital status (detailed) : Married                    | 7497  | 199.66           | 206.47          | [163.18, 248.00] |
| Race and ethnicity : Other or multiple, non-Hispanic   | 717   | 149.27           | 154.27          | [68.09, 229.77]  |
| Race and ethnicity : Black, non-Hispanic               | 2495  | 138.37           | 142.82          | [86.76, 199.90]  |
| Age group : 50-64                                      | 3760  | 202.35           | 208.84          | [167.97, 247.03] |
| Region : Northeast                                     | 2892  | 204.39           | 210.87          | [164.04, 257.12] |
| Race and ethnicity : White, non-Hispanic               | 10243 | 218.16           | 225.03          | [181.96, 267.59] |

| Domain                                                   | <i>n</i> | Scalar CATE (\$) | Locally smeared | Local 95% Cr     |
|----------------------------------------------------------|----------|------------------|-----------------|------------------|
| Poverty cat. : Low-income                                | 2458     | 126.79           | 130.77          | [72.21, 205.38]  |
| Sex : Female                                             | 9835     | 189.99           | 195.93          | [153.55, 238.20] |
| Overall : All                                            | 18763    | 188.02           | 193.76          | [160.26, 227.56] |
| Sex : Male                                               | 8928     | 185.86           | 191.37          | [156.99, 227.75] |
| Family income (weighted tertile) : Middle-income tertile | 5922     | 207.89           | 214.00          | [178.60, 249.27] |
| Poverty cat. : Middle-income                             | 5243     | 212.02           | 218.09          | [175.39, 262.89] |
| Region : South                                           | 7375     | 161.28           | 165.88          | [130.87, 203.88] |
| Age group : 0-17                                         | 3647     | 127.04           | 130.61          | [84.92, 184.24]  |
| Marital status : Not married                             | 11266    | 180.28           | 185.30          | [151.60, 221.13] |
| Region : West                                            | 4659     | 214.42           | 220.11          | [170.85, 270.78] |
| Race and ethnicity : Hispanic                            | 4165     | 144.82           | 148.15          | [115.04, 184.03] |
| Family income (weighted tertile) : Low-income tertile    | 7511     | 172.20           | 175.83          | [139.34, 217.24] |
| Marital status (detailed) : Divorced                     | 1933     | 226.30           | 230.70          | [147.61, 297.80] |
| Marital status (detailed) : Under 16                     | 3151     | 113.70           | 115.67          | [67.98, 172.64]  |
| Poverty cat. : Poor                                      | 2834     | 155.61           | 158.24          | [115.11, 202.15] |
| Marital status (detailed) : Widowed                      | 1261     | 244.47           | 247.56          | [136.59, 353.29] |
| Age group : 65+                                          | 4536     | 248.95           | 252.05          | [181.06, 315.55] |
| Poverty cat. : Near-poor                                 | 877      | 168.89           | 170.96          | [104.13, 257.55] |
| Marital status (detailed) : Separated                    | 342      | 143.67           | 145.08          | [79.51, 234.29]  |

Across all 33 reported levels (the overall sample plus the 32 subgroup levels), and on both the unweighted and the design-weighted footing, every locally smeared estimate exceeds its scalar counterpart, by 1.0 to 5.0% on the analytic sample and 1.1 to 5.2% on the population basis, with no sign change and a rank correlation of 0.999 between the two orderings. No covariate family changes its internal ordering, and the largest gap between levels widens rather than narrows, from \$135.3 to \$136.4 sample-side and from \$137.1 to \$137.8 population-side. The subgroup structure that carries the racial and ethnic gradient discussed above is therefore preserved under the more conservative re-transformation: non-Hispanic White remains the largest racial and ethnic effect at \$218.2 (locally smeared \$225.0, a 3.2% shift), and every locally smeared point estimate falls inside the scalar-smear credible interval it replaces. The subgroup conclusion is, if anything, marginally more exposed to residual-scale misscaling than the average effect (5.0% against 3.0%), but the exposure is bounded and small relative to the width of every interval involved.

**Comparison with the unadjusted association.** The unadjusted difference in mean dental expenditure between insured and uninsured respondents is \$185.9, or \$190.6 weighted. The adjusted dollar-scale effect is therefore essentially equal to the raw association rather than a fraction of it, and the earlier characterization of the estimate as roughly half the unadjusted gap does not survive recomputation. A linear regression of expenditure on coverage and the same covariates attenuates the raw gap to \$117.0, and the Gamma-hurdle benchmark reports \$147.9, so the absence of attenuation is specific to the flexible two-part fit rather than a property of the data alone.

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The attenuation that the selection story predicts does appear, but on the participation margin rather than on the dollar scale. The unadjusted difference in the probability of any dental spending is 0.201, and the adjusted participation-margin effect is 0.139 with a design-aware interval of  $[0.124, 0.155]$ , so roughly three tenths of the raw participation association is absorbed by the covariates. On the dollar scale almost none of it is. The natural reading is that the available covariates capture the demographic and economic correlates of coverage but not the latent oral-health demand that drives selection into it, so the dollar-scale estimate should be regarded as more exposed to residual confounding than its interval suggests. This is a limitation of the adjustment set rather than a property of the estimator, and it is carried forward into Section 6.6.

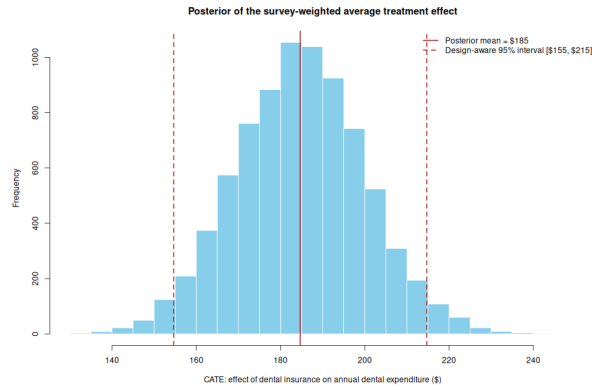


Figure 16: Posterior distribution of the survey-weighted ATE, with the design-aware 95% interval marked. The posterior concentrates near \$185 and lies entirely above zero.

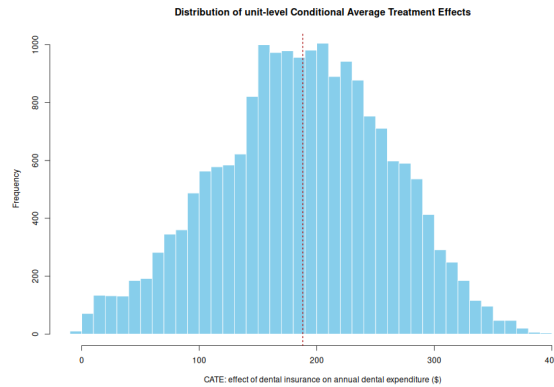


Figure 17: Marginal distribution of unit-level posterior-mean CATEs across all 18,763 individuals. This marginal view is shown for reference only; all inferential subgroup summaries use the draw-by-draw construction of Section 6.2.

**The ZIC-BCF-Smear Hurdle-Margin ATE** Although its primary estimand is the overall dollar-scale CATE, ZIC-BCF-Smear is a two-part model and therefore also delivers a participation (hurdle-margin) treatment effect from its probit hurdle stage. Let  $\mu_b(x)$  and  $\tau_b(x)$  denote the probit-scale prognostic and treatment-effect draws returned by the hurdle stage (`fit$mu_b` and `fit$tau_b`). The covariate-specific participation probabilities under control and treatment are  $p_0(x) = \Phi(\mu_b(x))$  and  $p_1(x) = \Phi(\mu_b(x) + \tau_b(x))$ , and the hurdle-margin CATE is  $p_1(x) - p_0(x)$ , formed draw-by-draw and then summarized exactly as the dollar-scale CATE. The overall hurdle-margin ATE is 0.139 as a sample average and 0.139 as a population target, with a design-aware 95% interval of [0.124, 0.155]: dental insurance raises the probability of any dental expenditure by about 14 percentage points. Reporting this margin makes explicit that the participation channel is a native output of ZIC-BCF-Smear and not only of the Gamma hurdle benchmark.

**Comparison with a Gamma Hurdle Benchmark** As a benchmark, the Bayesian Gamma hurdle specification of [Oganisian et al. \(2021\)](#) was fit on the same analytic sample, with the same design matrix, the same four chains, and the same weighting and variance treatment. This model recombines the participation and positive-intensity pieces draw-by-draw. The resulting response-scale ATE is \$147.87 as a sample average, with a 95% credible interval from \$90.06 to \$208.44, and \$143.20 as a population target with a design-aware interval from \$88.95 to \$197.45. Its hurdle-margin ATE is 0.118 (design-aware 95% interval [0.101, 0.134]), so it estimates that dental insurance raises the probability of any dental expenditure by about 11.8 percentage points. Its convergence is unproblematic, with  $\hat{R}$  at most 1.001 and effective sample sizes above 3,900. These ATE-only benchmark results show that the positive average effect is not an artifact of ZIC-BCF-Smear: the Gamma hurdle model also finds a credibly positive response-scale effect. The ZIC-BCF-Smear ATE is nevertheless larger and much more tightly concentrated, with a posterior standard deviation of \$15.7 against \$27.6, consistent with the simulation finding that flexible smearing-based re-transformations are less fragile under heavy right skew and positive-intensity misspecification than a parametric Gamma positive-intensity likelihood.

### Full Subgroup CATE Table

Table 16 reports every subgroup CATE with both estimands and their intervals. All subgroups have posterior probability of a positive effect equal to 1.00. That uniformity is a statement about location rather than a sign of understated uncertainty: the overall effect of about \$185 is roughly twelve posterior standard deviations from zero, and no subgroup estimate falls below \$124, so a posterior mass below zero would be surprising. The informative question is therefore not whether each subgroup effect is positive but whether subgroups differ, and that question is answered by the contrasts below, of which 37 of 56 are not credible. The convergence diagnostics reported above establish separately that these probabilities are not artifacts of a short or poorly mixed chain. Figure 18 plots the same estimates against the overall ATE.

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Table 16:: Subgroup CATE estimates (dollars), MEPS 2023. The sample column is the unweighted analytic-sample average with its 95% credible interval; the population column is the survey-weighted target with its design-aware 95% interval.

| Covariate                        | Subgroup                        | <i>N</i> | Analytic-sample average | Population      |
|----------------------------------|---------------------------------|----------|-------------------------|-----------------|
| Overall                          | All                             | 18,763   | 188.02 [156.95, 220.67] | 184.67 [154.63] |
| Sex                              | Male                            | 8,928    | 185.86 [153.06, 220.98] | 182.56 [149.75] |
| Sex                              | Female                          | 9,835    | 189.99 [149.59, 230.49] | 186.70 [148.40] |
| Race and ethnicity               | Hispanic                        | 4,165    | 144.82 [113.97, 178.04] | 146.57 [114.08] |
| Race and ethnicity               | White, non-Hispanic             | 10,243   | 218.16 [176.61, 260.39] | 208.95 [170.75] |
| Race and ethnicity               | Black, non-Hispanic             | 2,495    | 138.37 [84.47, 192.74]  | 137.53 [84.85]  |
| Race and ethnicity               | Asian, non-Hispanic             | 1,143    | 208.13 [152.57, 275.42] | 203.77 [143.71] |
| Race and ethnicity               | Other or multiple, non-Hispanic | 717      | 149.27 [64.24, 223.36]  | 141.60 [64.16]  |
| Age group                        | 0–17                            | 3,647    | 127.04 [83.87, 176.92]  | 126.96 [80.03]  |
| Age group                        | 18–34                           | 3,352    | 169.06 [139.78, 201.82] | 174.18 [141.62] |
| Age group                        | 35–49                           | 3,468    | 175.27 [147.15, 205.33] | 178.22 [148.39] |
| Age group                        | 50–64                           | 3,760    | 202.35 [164.11, 239.59] | 207.68 [168.99] |
| Age group                        | 65+                             | 4,536    | 248.95 [176.24, 315.74] | 250.26 [181.53] |
| Poverty category                 | Poor                            | 2,834    | 155.61 [114.17, 198.17] | 153.07 [113.13] |
| Poverty category                 | Near-poor                       | 877      | 168.89 [103.54, 252.18] | 165.63 [96.64]  |
| Poverty category                 | Low-income                      | 2,458    | 126.79 [69.09, 198.41]  | 124.00 [62.16]  |
| Poverty category                 | Middle-income                   | 5,243    | 212.02 [171.07, 255.50] | 203.08 [163.92] |
| Poverty category                 | High-income                     | 7,351    | 206.16 [159.17, 250.69] | 198.62 [156.75] |
| Family income (weighted tertile) | Low tertile                     | 7,511    | 172.20 [136.59, 212.72] | 168.99 [133.05] |
| Family income (weighted tertile) | Middle tertile                  | 5,922    | 207.89 [173.71, 242.16] | 201.35 [168.87] |
| Family income (weighted tertile) | High tertile                    | 5,330    | 188.25 [144.99, 230.87] | 183.65 [143.18] |
| Region                           | Northeast                       | 2,892    | 204.39 [159.66, 249.48] | 196.76 [155.52] |
| Region                           | Midwest                         | 3,837    | 195.04 [149.37, 247.82] | 187.86 [142.05] |
| Region                           | South                           | 7,375    | 161.28 [128.68, 196.40] | 164.84 [131.46] |
| Region                           | West                            | 4,659    | 214.42 [166.67, 265.17] | 205.96 [160.39] |
| Marital status                   | Married                         | 7,497    | 199.66 [159.35, 239.49] | 197.32 [159.09] |
| Marital status                   | Not married                     | 11,266   | 180.28 [147.94, 214.96] | 176.26 [144.20] |

Marital status is also reported at its full resolution, since the binary grouping pools respondents under 16 with never-married adults. On the detailed grouping the married, widowed, divorced and never-married levels all exceed the under-16 level credibly, at \$84.20, \$136.32, \$112.96 and \$80.43 respectively, which is the age gradient reappearing through a variable whose inapplicable category is defined by age. No contrast among the adult marital levels is credible.

### Between-Subgroup Contrasts

A difference in point estimates is meaningful only if it survives posterior uncertainty, and on a complex survey it should also survive the sampling design. Because subgroup effects are computed draw-by-draw, the posterior of any contrast is available directly,

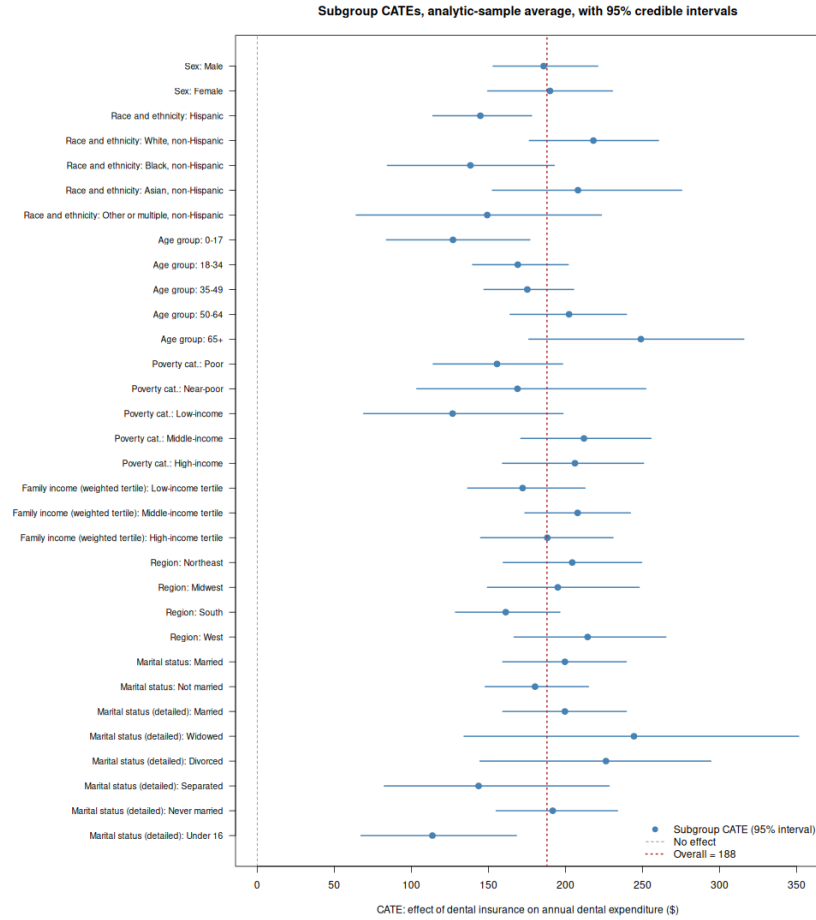


Figure 18: Subgroup CATEs with 95% credible intervals on the analytic-sample average. The dashed line marks no effect and the dotted line marks the overall ATE.

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and the design-based variance of a contrast is obtained from the differenced linearized values of Section 6.2. Of the 56 pairwise contrasts, 16 exclude zero on the analytic-sample average and 18 on the design-aware population target. The contrasts that are credible under the design-aware criterion are the following, with population-target estimates and intervals.

On race and ethnicity, non-Hispanic White beneficiaries gain \$71.42 [\$11.96, \$130.87] more than non-Hispanic Black beneficiaries, and Hispanic beneficiaries gain \$62.38 [−\$100.02, −\$24.74] less than non-Hispanic White beneficiaries and \$57.20 [−\$110.33, −\$4.06] less than non-Hispanic Asian beneficiaries. On age, the effect rises monotonically and seven of the ten pairwise contrasts are credible, the largest being children versus the 65-and-over group at −\$123.30 [−\$204.88, −\$41.73]. On income and poverty, the low-income poverty category gains credibly less than both the middle-income category, at −\$79.08 [−\$145.02, −\$13.14], and the high-income category, at −\$74.62 [−\$146.85, −\$2.38], and on the weighted income tertiles the low tertile gains less than the middle tertile at −\$32.36 [−\$59.53, −\$5.19]. On the detailed marital grouping, the four adult levels each exceed the under-16 level as noted above.

Two results reported as credible in the first version of this study do not survive the corrections, and the change is reported rather than absorbed. The South-versus-West regional contrast is −\$53.14 [−\$102.69, −\$0.51] on the analytic-sample average, which excludes zero, but −\$41.12 [−\$87.62, \$5.38] on the design-aware population target, which does not. Since a regional claim about the United States is precisely a population claim, the design-aware column is the relevant one and the regional gradient should now be described as suggestive rather than credible. It was in any case the least literature-anchored of the reported gradients. The White-versus-Asian and White-versus-other-race contrasts are likewise no longer credible, which is a direct consequence of removing 3,758 Hispanic respondents from the White category and of estimating a non-Hispanic Asian effect that is high rather than low.

Sex shows no credible heterogeneity on either estimand (male minus female = −\$4.14 [−\$42.76, \$34.49] on the population target), and neither does the binary married-versus-not-married contrast on the dollar scale. Figure 19 displays all pairwise contrasts with the credible ones highlighted, Figure 20 shows the posterior subgroup densities for race and ethnicity and for age, and Figure 21 plots unit-level posterior-mean CATEs against the two continuous covariates.

### Hurdle-Margin (Participation) Subgroup Effects

The hurdle-margin CATE of Section 6.3 is summarized by subgroup exactly as the dollar-scale CATE, averaging the unit-level participation effects  $p_1(X_i) - p_0(X_i)$  within each posterior draw. Table 17 reports every subgroup hurdle-margin CATE on the survey-weighted population target, in probability points, with its design-aware 95% interval; all subgroups have posterior probability of a positive participation effect equal to 1.00.

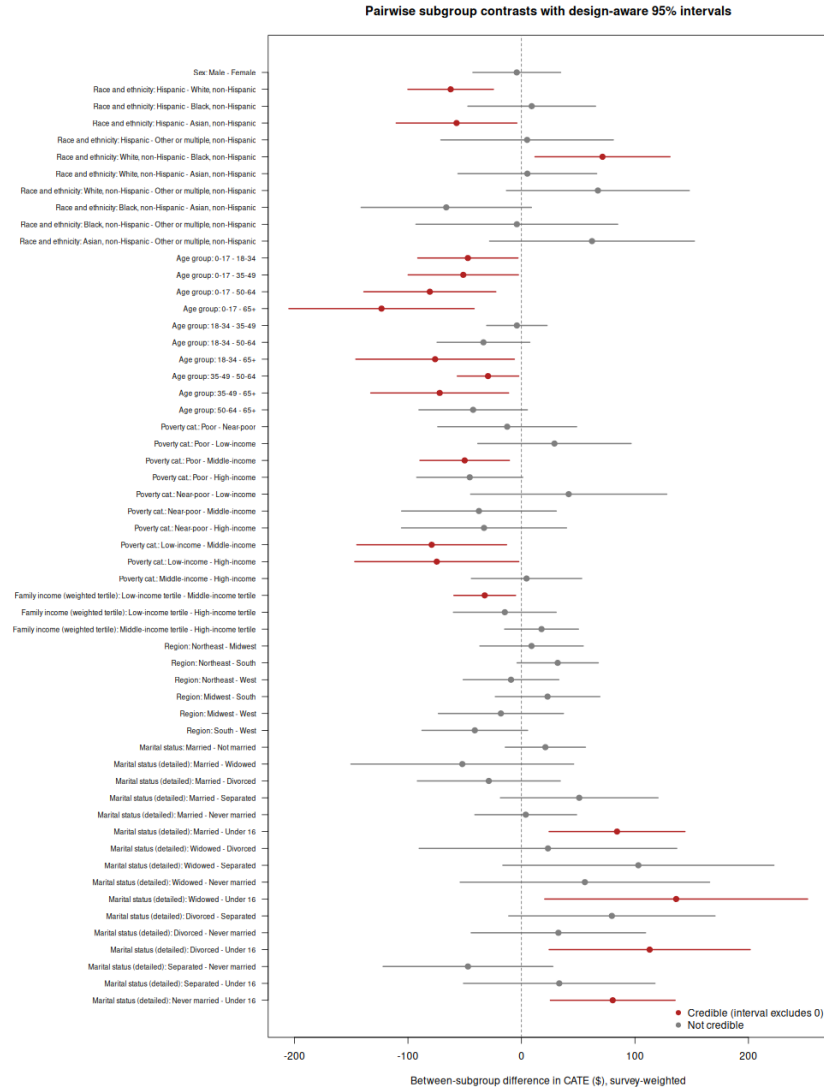


Figure 19: All pairwise between-subgroup CATE contrasts on the survey-weighted population target, with design-aware 95% intervals; contrasts whose interval excludes zero are highlighted.

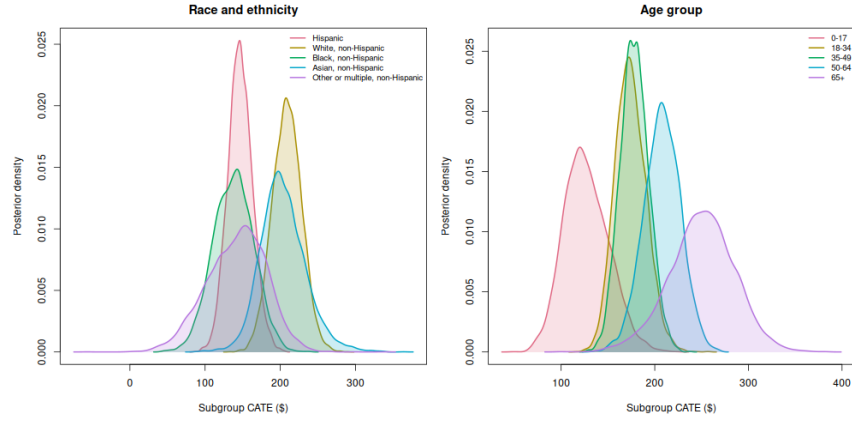


Figure 20: Posterior densities of the survey-weighted subgroup CATEs for race and ethnicity and for age. The heterogeneity is visible as a separation of whole posterior distributions, not merely of point estimates.

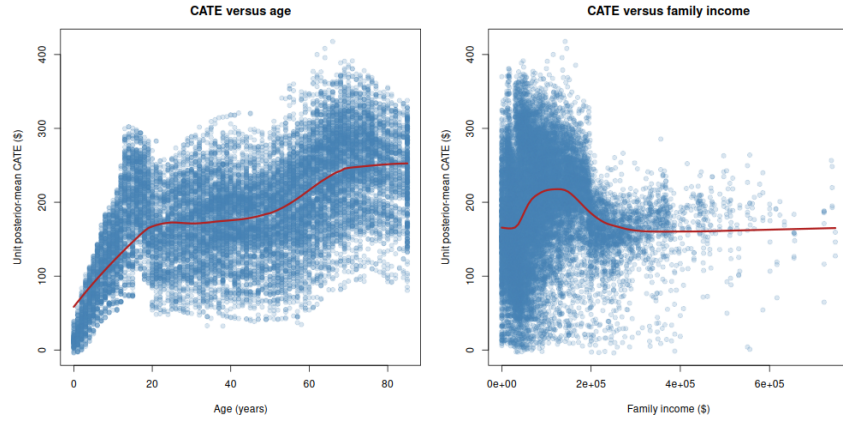


Figure 21: Unit-level posterior-mean CATE against age and family income, with a LOWESS trend. The insurance effect rises steadily with age and increases with income before flattening at the top of the distribution, consistent with the subgroup tables.

Table 17:: ZIC-BCF-Smear hurdle-margin (participation) subgroup CATEs, in probability points, on the survey-weighted population target with design-aware 95% intervals, MEPS 2023. Entries are the increase in the probability of any dental expenditure attributable to dental insurance.

| Covariate                        | Subgroup                        | <i>N</i> | Population target    |
|----------------------------------|---------------------------------|----------|----------------------|
| Overall                          | All                             | 18,763   | 0.139 [0.123, 0.155] |
| Sex                              | Male                            | 8,928    | 0.143 [0.121, 0.165] |
| Sex                              | Female                          | 9,835    | 0.136 [0.113, 0.158] |
| Race and ethnicity               | Hispanic                        | 4,165    | 0.129 [0.103, 0.155] |
| Race and ethnicity               | White, non-Hispanic             | 10,243   | 0.151 [0.130, 0.171] |
| Race and ethnicity               | Black, non-Hispanic             | 2,495    | 0.109 [0.063, 0.154] |
| Race and ethnicity               | Asian, non-Hispanic             | 1,143    | 0.152 [0.114, 0.190] |
| Race and ethnicity               | Other or multiple, non-Hispanic | 717      | 0.098 [0.027, 0.168] |
| Age group                        | 0–17                            | 3,647    | 0.093 [0.057, 0.128] |
| Age group                        | 18–34                           | 3,352    | 0.151 [0.130, 0.172] |
| Age group                        | 35–49                           | 3,468    | 0.154 [0.135, 0.173] |
| Age group                        | 50–64                           | 3,760    | 0.156 [0.137, 0.175] |
| Age group                        | 65+                             | 4,536    | 0.148 [0.125, 0.172] |
| Poverty category                 | Poor                            | 2,834    | 0.135 [0.105, 0.165] |
| Poverty category                 | Near-poor                       | 877      | 0.142 [0.090, 0.194] |
| Poverty category                 | Low-income                      | 2,458    | 0.080 [0.035, 0.124] |
| Poverty category                 | Middle-income                   | 5,243    | 0.157 [0.134, 0.180] |
| Poverty category                 | High-income                     | 7,351    | 0.145 [0.123, 0.167] |
| Family income (weighted tertile) | Low tertile                     | 7,511    | 0.130 [0.109, 0.150] |
| Family income (weighted tertile) | Middle tertile                  | 5,922    | 0.153 [0.135, 0.172] |
| Family income (weighted tertile) | High tertile                    | 5,330    | 0.135 [0.110, 0.160] |
| Region                           | Northeast                       | 2,892    | 0.142 [0.121, 0.163] |
| Region                           | Midwest                         | 3,837    | 0.141 [0.117, 0.166] |
| Region                           | South                           | 7,375    | 0.132 [0.113, 0.151] |
| Region                           | West                            | 4,659    | 0.147 [0.122, 0.172] |
| Marital status                   | Married                         | 7,497    | 0.154 [0.135, 0.174] |
| Marital status                   | Not married                     | 11,266   | 0.129 [0.111, 0.148] |

Eleven of the 56 participation-margin contrasts are credible on the design-aware criterion. Children gain credibly less than every older group (for example  $-0.059$  [ $-0.098, -0.020$ ] against the 18–34 group and  $-0.063$  [ $-0.103, -0.023$ ] against the 50–64 group); the low-income poverty category gains less than the middle-income and high-income categories, at  $-0.077$  [ $-0.127, -0.027$ ] and  $-0.065$  [ $-0.116, -0.015$ ], and the low income tertile less than the middle at  $-0.024$  [ $-0.040, -0.007$ ]; and married respondents gain more than unmarried ones at  $+0.025$  [ $0.004, 0.046$ ].

The two margins are internally coherent on age and income, which reproduce the same ordering on both scales. They part company on race and ethnicity: the point estimates order the groups the same way, with non-Hispanic White and non-Hispanic Asian above Hispanic and non-Hispanic Black, but on the participation scale no racial

or ethnic contrast is credible, whereas on the dollar scale three are. The reading is that the racial and ethnic gradient in this application lives mainly in how much is spent conditional on spending anything, rather than in whether coverage induces a visit at all. This differs from what the first version of this study reported, where a credible White-minus-Black participation contrast appeared; that contrast was estimated on a White category containing 3,758 Hispanic respondents and on an unweighted basis, and it does not survive either correction.

## 6.4 Reconciliation with the Applied Literature

The applied estimates are read against the dental health-economics literature dimension by dimension. Broadly, the *direction* of every effect converges with prior work; the novel contribution is translating well-documented *associational* gradients into *credible subgroup causal* effects on the raw dollar scale, with direct posterior contrasts.

**Average effect.** A positive ATE converges strongly with prior MEPS evidence that coverage predicts more visits and higher expenditure (Manski et al., 2002; Christian and Chattopadhyay, 2014; Naib et al., 2026). The finding sits naturally within the broad evidence that dental care imposes the highest financial barriers among health services, so that lowering the point-of-service price causally unlocks deferred care (Vujicic et al., 2016). The relationship to the standard selection-plus-enabling decomposition (Manski et al., 2015; Meyerhoefer et al., 2014) is more equivocal than the first version of this study reported, for the reason set out above: on the dollar scale the adjusted estimate is essentially the unadjusted gap, so these covariates absorb none of the presumed selection, whereas on the participation margin about three tenths of the raw association is absorbed. The decomposition is therefore supported on the participation channel and not demonstrated on the intensity channel.

**Race and ethnicity.** Larger gains for non-Hispanic White than for non-Hispanic Black and Hispanic enrollees converge with the extensive literature on racial and ethnic oral-health disparities, in which minority populations report lower dental-visit rates and lower expenditures (Zhang, 2016; Reda et al., 2018; Robison et al., 2020). The causal reading is sharper than the descriptive one: providing identical coverage does not yield identical spending increases across groups, pointing to non-financial barriers, such as provider supply, oral-health literacy, and differences in treatment recommendations, that coverage alone does not remove (Reda et al., 2018; Yu, 2025). One result cuts across the simple ordering and is reported as it stands: non-Hispanic Asian enrollees have the second-largest effect, at \$205.0, close to the non-Hispanic White estimate and credibly above the Hispanic estimate. The first version of this study placed Asian enrollees low, at \$154, but did so using a race-only variable that assigned Hispanic respondents to other categories and on an unweighted basis, so the earlier ordering is not evidence against the present one. The disparities literature is itself less settled for Asian American subpopulations than for Black and Hispanic ones, and aggregation across highly heterogeneous subgroups is a known limitation of the single MEPS category (Yu, 2025); we therefore treat this estimate as a finding to be reported rather than as a contradiction to be explained away.

**Age.** The monotone rise of the effect with age, from \$127 among children to \$249 among those aged 65 and over, converges with evidence that dental expenditure increases with age and that older adults face particular coverage gaps (Christian and Chattopadhyay, 2014; Zhang, 2016). The pattern is economically intelligible: pediatric care is dominated by low-cost preventive services and public coverage, whereas older adults carry a heavier restorative burden and, lacking routine coverage under traditional Medicare, exhibit large latent demand that coverage unlocks (Manski et al., 2015; Meyerhoefer et al., 2014). Estimating this gradient as a causal heterogeneity, rather than a spending level, is largely new. It is also the most robust result in this study, surviving both the encoding correction and the survey-design correction with five of ten pairwise contrasts credible.

**Income and poverty.** Smaller effects for low-income than for middle-income and high-income enrollees converge with the repeated finding that lower-income groups use less dental care and face more barriers (Manski et al., 2002; Reda et al., 2018; Li and Yao, 2021; Yu, 2025). The mechanism emphasized in the financing literature is that dental insurance carries substantial cost-sharing and low annual caps, so coverage is a weaker enabling resource for those least able to absorb out-of-pocket costs (Vujicic et al., 2016). The gradient is not monotone across the five poverty categories, since the poor and near-poor categories sit above the low-income category on both margins, which is consistent with public coverage being most complete at the bottom of the distribution and thinnest just above it.

**Region.** A smaller effect in the South than the West remains visible in the point estimates, at \$165.5 against \$205.2, and geography is an established predictor of dental spending and utilization, with rural and Southern populations showing lower use (Christian and Chattopadhyay, 2014; Luo et al., 2021; Joudi et al., 2024). The contrast no longer excludes zero once the sampling design is respected, however, so it is reported as suggestive rather than credible. Insurance-effect heterogeneity by U.S. region is rarely estimated, and this study does not establish it.

**Null heterogeneity.** The absence of credible heterogeneity by sex modestly diverges from utilization studies, which often find women use more dental care (Reda et al., 2018; Zhang, 2016); the reconciliation is that a higher female utilization frequency need not translate into a larger dollar increment from coverage. The absence of dollar-scale marital-status heterogeneity among adults is broadly consistent with the literature, in which marital status appears in utilization models but is not a dominant determinant of spending differences (Zardak et al., 2023; Reda et al., 2018).

**Methodological placement.** Finally, the two-part and zero-inflated modeling choices are themselves well grounded: semicontinuous health-service outcomes routinely require hurdle or two-part treatments (Feng, 2021; Neelon et al., 2016), and the retransformation problem that Duan's smearing addresses is the classic obstacle to reporting

log-scale models on the dollar scale (Duan, 1983; Manning and Mullahy, 2001). ZIC-BCF-Smear extends this lineage by making both the participation and intensity surfaces fully nonparametric and by carrying full posterior uncertainty onto the dollar scale.

### **ZIC-BCF-Smear versus the Gamma Hurdle benchmark on literature convergence.**

Read on the *dollar-scale* subgroup CATE, ZIC-BCF-Smear converges to the dental health-economics literature and the Gamma Hurdle benchmark of Oganisian et al. (2021) does not. Both estimators place the overall response-scale ATE credibly above zero, but they split on the heterogeneity question, which is the level at which the literature is most informative, and the split is stark under design-aware inference: ZIC-BCF-Smear returns 19 credible dollar-scale contrasts out of 56, and the Gamma Hurdle returns one.

That single contrast runs against the literature rather than with it, placing the low income tertile \$124.06 [\$2.80, \$245.43] *above* the high tertile, the reverse of the cost-sharing mechanism that the financing literature describes (Vujicic et al., 2016; Reda et al., 2018). Its race and ethnicity point estimates do not order the groups as three decades of applied work do, at \$144.2 for non-Hispanic Black, \$143.3 for non-Hispanic White and \$138.4 for Hispanic enrollees, differences far smaller than their intervals. Its dollar-scale age gradient is non-monotone, rising to \$166.9 at ages 18–34, falling to \$124.0 at ages 50–64, and rising again to \$176.0 at 65 and over. And its regional ordering is inverted relative to the utilization literature, placing the South highest at \$198.7 and the Northeast lowest at \$81.4.

What rules out the reading that the Gamma Hurdle is simply measuring something different is that it recovers credible structure on its participation margin, where its decomposition is identically suited, returning 17 credible contrasts to ZIC-BCF-Smear’s 11. The benchmark loses the heterogeneity precisely where its distributional assumption binds. The mechanism is the one identified in the link-nonlinear simulation results of the main manuscript: the Gamma positive-intensity component imposes a single parametric tail shape that smooths away the heavy right tail in which the literature-aligned heterogeneity lives, whereas the nonparametric smearing re-transformation absorbs it. On the dollar CATE, where ZIC-BCF-Smear reports its primary estimand, the practical consequence is that the literature-aligned contrasts survive posterior and design uncertainty under ZIC-BCF-Smear but not under Gamma Hurdle.

## **6.5 Policy Reading of the Heterogeneity**

Interpreted causally, the estimated heterogeneity carries a cautionary policy implication that a naive equity argument for coverage expansion can miss. Because the marginal dollar effect of coverage is systematically larger for already-advantaged subgroups (non-Hispanic White, older, and higher-income enrollees), a uniform expansion of dental insurance could, in absolute terms, *widen* rather than compress spending gaps, since insurance functions as an enabling resource whose value depends on complementary structural conditions (ability to absorb cost-sharing, provider availability, favorable fee and scope-of-practice environments) that disadvantaged groups disproportionately lack

(Vujicic et al., 2016; Manski et al., 2015; Reda et al., 2018). The corresponding implication is that coverage expansion is most equity-enhancing when paired with measures that relax those frictions: reduced cost-sharing for low-income enrollees, higher public-program reimbursement to raise provider participation, and supply-side reforms in under-served areas (Yu, 2025; Joudi et al., 2024).

Two qualifications bound how far this reading can be pushed. It rests on the survey-weighted column of the tables above, which is the only column that supports a national statement; the regional element of the earlier version of this argument has been withdrawn because it does not survive design-aware inference. And these are policy conjectures generated by the estimated causal heterogeneity rather than tests performed here, offered to motivate targeted follow-up analyses.

## 6.6 Limitations of the Applied Study

Several caveats bound the interpretation.

First, the analysis is *observational*: the reported effects are causal only under conditional ignorability of the coverage–spending relationship given  $X$  and under the subpopulation strong-ignorability condition (Assumption 6.1). This is the binding limitation, and the comparison with the unadjusted association sharpens it. Dental coverage is plausibly subject to selection on latent oral-health demand, and the covariates available here absorb about three tenths of the raw participation association but essentially none of the raw dollar-scale association. That is more consistent with an adjustment set that misses the relevant latent demand than with an absence of selection, so the dollar-scale estimate should be regarded as more exposed to residual confounding than its interval conveys.

Second, MEPS 2023 is a *cross-section*, so the effects are contemporaneous associations adjusted for confounding rather than dynamic responses to a coverage change, and expenditure is measured with survey and recall error.

Third, the treatment is a *binary coverage indicator* that collapses heterogeneous plan generosity (caps, coinsurance, networks); the estimated effect is an average over this unmodeled plan heterogeneity.

Fourth, the *survey design is handled outside the model*. Weighting acts on the reported linear functionals of unit-level effects, and the design-based variance is added to the posterior variance under an approximate independence assumption, but the design does not influence the fitted response surface itself, because the sampler has no weight interface. A design-informed prior would be the principled alternative and is not attempted here. Related to this, 443 analytic records carry a zero person weight and therefore contribute to the model fit but not to any population estimate.

Fifth, some subgroups are small, notably non-Hispanic other or multiple race ( $N = 717$ ) and the separated marital category ( $N = 342$ ), so their CATEs and contrasts carry wider intervals and should be read cautiously.

Sixth, the continuous stage assumes *homoskedastic log-scale errors*, and in this application that assumption is decisively rejected, at  $p = 1.2 \times 10^{-24}$  with residual standard

deviation rising monotonically in the fitted mean. The locally smeared sensitivity analysis shows the average effect moving by only 3.1%, so the reported estimate is not carried by the assumption, but the calibration of subgroup credible intervals under strong mean-variance dependence is not guaranteed by that check alone.

Finally, the *subgroup comparisons are not multiplicity-adjusted*. Fifty-six pairwise contrasts are examined on each of two margins, and while the pattern of credible contrasts is coherent across margins and consistent with prior literature, individual contrasts near the boundary should be read accordingly.

These limitations motivate longitudinal and plan-resolved follow-up work rather than qualifying the central finding of a credibly positive and heterogeneous causal effect of dental coverage on dental spending.

## 6.7 Reproducibility

This study reproduces in full from a clean checkout of the analysis repository accompanying the `zicbcf` package (Souto, 2026), and all paths below are relative to its `applied_study/` directory. The MEPS public-use files are released by the Agency for Healthcare Research and Quality and are freely redistributable, subject only to two conduct restrictions that survive redistribution: the data may be used for statistical reporting and analysis only, and they may not be linked with individually identifiable records from any source other than MEPS itself and the National Health Interview Survey from which the MEPS sample is drawn.

The raw HC-251 file is fetched and prepared by `download_meps_h251.R`, which is committed; the resulting `h251.csv` is excluded from version control only because it is roughly 60 MB, and the script reproduces it byte-comparably on the analytic variables. The cleaning rules, design matrix, subgroup definitions, weighted-quantile helper and survey-design variance functions are defined once in `meps_common.R` and are shared by every fit. The ZIC-BCF-Smear analysis is run by `run_study.R`, which writes the posterior summaries, the convergence diagnostics and the smearing diagnostics; the subgroup tables, contrasts and figures are produced without refitting by `run_cate_subgroups.R`. The Gamma-hurdle benchmark is run by `run_gamma_hurdle.R` and summarized by `run_gamma_hurdle_subgroups.R`. A README in the same directory documents the run order and the two reported estimands.

The repository previously excluded every derived results table and every markdown file in this directory through two blanket ignore patterns, which made the MEPS arm irreproducible for no benefit, since none of that material is restricted. Those patterns have been narrowed to the raw input file, the regenerable posterior-draw objects and the local manuscript build materials, so every derived table, figure and note is now committed. The convergence diagnostics for the simulation grid are produced by `simulation_studies/run_convergence_diagnostics.R`, and the link-linear generators it needs are available separately from the full simulation driver in `simulation_studies/dgps_linear.R`.

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